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Structure–Property Relationships of Solid Oxide Fuel Cell Electrodes Using Real and Reconstructed Microstructures

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ABSTRACT

Solid oxide fuel cells (SOFCs) are a promising technology for producing clean energy in the future. To enhance the efficiency and the lifetime of SOFCs and SOFC stacks, novel materials and innovative material compositions for the constituent layers—namely the cathode, anode, and electrolyte—must be developed. Since the electrodes are of tremendous importance for an efficient and long-term operation, computational methods are required to enable and accelerate the development process. In this work, the microstructure of an SOFC anode is characterised by certain geometrical descriptors, such as the volume fraction, the two-point correlation function, the lineal-path function, the tortuosity, the percolation of each constituent, and the triple phase boundary (TPB). In addition, the effective physical properties are determined by the first-order computational homogenisation method to enable the usage of real anisotropic material parameters in macroscopic fuel cell simulations. Special attention is paid to the mechanical behaviour, including secondary creep. A general framework for creating structure-property relationships for fuel cell electrodes is established. Since it is unfeasible to build large datasets based on experimental tomography images, the microstructure characterisation and reconstruction tool *MCRpy* is used to generate microstructures based on the geometrical descriptors (i) of the whole three-dimensional tomography image and (ii) of only three two-dimensional slices. A two-step approach is employed with a differentiable microstructure characterisation and reconstruction (DMCR) algorithm and the classical Yeong-Torquato algorithm. The results show a very good agreement of the geometrical and the physical properties between the original tomography image and the reconstructed microstructures. However, to decrease the small differences in the properties, it is necessary to include further geometrical descriptors into the reconstruction process in further research. The created framework can support the development of optimised electrode microstructures using inverse design methods.

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1 | Introduction

Solid oxide fuel cells (SOFCs) are able to directly convert chemical energy into electrical energy in an efficient and environmentally clean manner [1]. A SOFC consists of an electrolyte that is sandwiched between two porous electrodes, namely the cathode and the anode, as well as bipolar interconnect and sealant materials. The fuel and the air flow through the gas channels and into the porous media towards the electrolyte. The reduction, at which oxygen is reduced to oxygen ions, occurs at the electrochemically active phase boundary, for example, at the double phase boundary when common mixed ionic-electronic conducting materials (e.g., lanthanum strontium cobalt ferrite) are utilised [2]. The oxygen ions are conducted by a dense ceramic electrolyte to the anode, where oxidation takes place. Using a commonly employed material composition of nickel (Ni) and yttrium-stabilised zirconium (YSZ) [2], the fuel is oxidised to protons, electrons, and a waste product. If, for instance, pure hydrogen is applied, the redox reactions are as follows:



Even though SOFC stacks are currently tried to be implemented in combined heat and power systems [3] due to their high total efficiency up to 80% [4], a broad commercial breakthrough of this technology is still hindered due to their limited lifetime [5]. Because of their high operating temperatures between 600°C to 1000°C and differing thermal expansion coefficients of each layer, high-temperature gradients arise within the fuel cell, which results in thermomechanical stresses [6, 7]. These thermomechanical stresses originate from manufacturing processes, the fuel cell design, or various stages of fuel cell operation [8]. In addition, inelastic deformations such as the creep behaviour are expected to significantly affect the long-term durability of each layer [9, 10]. Therefore, it is inevitable to also consider the complex mechanical behaviour in macroscopic fuel cell simulations in order to make lifetime predictions [11].

The electrodes of the SOFC are known to have a substantial influence on the overall performance and on the durability [12–15]. For instance, it is required that the cathode and the anode have high electric and ionic conductivities and a suitable permeability such that the flow of fuel and air is fast. Furthermore, they are supposed to exhibit good morphological and mechanical stability, which is in contrast to a high pore volume fraction [5]. In addition, high thermal conductivities are beneficial to significantly decrease thermal temperature gradients and thus stresses [16]. Creep effects are important because tensile stresses are relieved in the cathode layer [17] and the anode layer [16], and other components of the fuel cell can also be affected by stress relaxation.

However, to accurately capture the influence of the electrodes in the entire fuel cell, the microstructure of these layers must first be known. To gain geometrical information about the electrodes, focused ion beam–scanning electron microscopy (FIB-SEM) is typically employed to characterise various microstructural features that are crucial for the SOFC operation. In this process, a series of two-dimensional images is taken by SEM. Between the records, the FIB ensures the removal of material. This combined procedure is therefore capable of precise milling and high-resolution imaging [18]. Various researchers determined both the morphological characteristics and the physical properties of the experimentally obtained three-dimensional (3D) microstructures [19–24].

The optimisation of anodes and cathodes is, however, not feasible by real-world experiments, as creating a large database, for example, data-driven models in the fuel cell application [25], would be too time-consuming and expensive. It is therefore convenient to use numerical tools for the fuel cell development and their constituents, as simulations are cheaper and faster than experimental studies in a laboratory. Many research groups investigated different methods to artificially generate and reconstruct 3D, realistic microstructures of SOFC electrodes. In [26, 27], a certain artificial domain is filled with spherical particles that are adjusted during the process to ensure connectivity between the spheres and phases of the material or to achieve a desired physical property for the composite material, respectively. Machine learning models, in particular generative adversarial networks, are based on large-scale and experimentally captured 3D microstructures and are applied to SOFC electrodes [28, 29]. In addition, there exists a variety of approaches to generate 3D microstructures from a single 2D image with the help of the Monte Carlo methodology [30], co-occurrence correlation functions [31], statistics-informed neural networks [32], and the denoising diffusion approach [33]. Further studies address approaches such as cellular automata [34] or Pluri-Gaussian methods [35] to find optimised morphologies for SOFC electrodes based on artificially generated microstructures. In the present work, a generalised framework to reconstruct random heterogeneous media is applied, which was developed in a previous work [36] and is already validated for effective mechanical properties of a binary β -Ti/TiFe alloy [37]. More specifically, the microstructure characterisation is based on certain

translation-invariant descriptors. The reconstruction is based on these descriptors and relies on an optimisation problem solved by a differentiable optimiser. Fewer iterations are required by this optimiser than by the Yeong-Torquato algorithm. This is implemented in the open-source tool *MCRpy* [38]. This tool is able (i) to generate periodic microstructures from aperiodic tomography images, which is beneficial for the computational homogenisation procedure, (ii) to obtain realistic 3D structures from 2D images, (iii) to create multiple similar realisations of one example and (iv) to interpolate in the descriptor space to generate, for example, SOFC electrodes with grain sizes that are currently not used in the manufacturing process. In particular, the ability to generate microstructures on modified geometrical descriptors is very valuable for developing and virtually testing new materials. For the reconstruction procedure in this work, *MCRpy* was extended by implementing new descriptors, such as the triple phase boundary (TPB) density and the extension of the descriptors to 3D. In prior applications, the descriptors were computed for each slice of each plane. However, 3D descriptors are used in the second step of the reconstruction, since the TPB cannot be clearly calculated in a 2D setting. In addition, new pixel selection probabilities are implemented to increase the efficiency of the Yeong-Torquato approach. Furthermore, *MCRpy* is validated for multi-physical/transport properties in contrast to purely mechanical quantities.

This work aims to create a basis for the development of SOFC electrodes. With the presented framework, structure-property relationships can be generated that help to find suitable materials, material compositions, as well as microstructure morphologies to increase both the efficiency and the lifetime of SOFCs. Therefore, important microstructural features are analysed and multi-physical and anisotropic effective properties are numerically determined based on a representative volume element ($\mathcal{RVE}$). The pipeline for the simulation chain is summarised in Figure 1 and explained in detail in the following sections.

Although there exist approaches to generate SOFC electrodes numerically, many of these methods rely on stochastic approaches, use only simplified descriptors, lack computational efficiency, do not consider real microstructural samples, or require more data to train neural networks. In contrast, the combination of DMCR and the Yeong-Torquato procedure is able to generate arbitrary and complex microstructures using only three 2D or one 3D sample as a reference, and requires only a small number of iterations, particularly compared to the widely used simulated annealing approach. The reconstruction approach can help to provide multiple microstructures, increasing the number of samples available for enhancing the microstructural design of SOFC electrodes. The results show that the FIB-SEM procedure can be replaced or supplemented by taking only three 2D images or searching for more suitable electrodes by varying the morphological descriptors. Overall, the framework for creating structure-property relationships can be used to optimise the electrode microstructure with respect to electrochemical performance (high power) and mechanical stability/durability.

The present paper is structured as follows: In Section 2, the descriptors that are important for characterising the morphology of the microstructure are presented. In Section 3, the homogenisation to determine the effective properties, including various conductivities, the fluid permeability, and the mechanical behaviour, is explained. Special attention is paid to the determination of the effective creep parameters. Besides, the anisotropy of the effective physical tensors is considered.

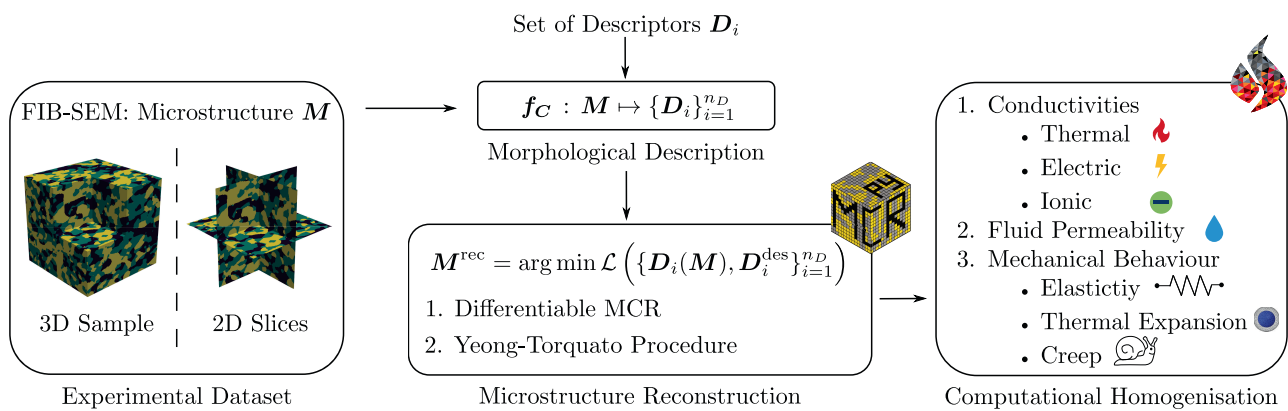


FIGURE 1 | Pipeline of the proposed simulation chain: Experimental focused ion beam–scanning electron microscopy (FIB-SEM) images as input data (either 2D or 3D) is the starting point for morphologically characterising the microstructure. The two-step optimisation procedure uses a set of descriptors to reconstruct the microstructure via (i) differentiable microstructure characterisation and reconstruction (DMCR) and (ii) the Yeong-Torquato approach. The original and reconstructed microstructures are meshed by finite elements, and the determination of certain effective properties is conducted by computational homogenisation.

Section 4 briefly represents both the differentiable algorithm for characterising and reconstructing microstructures and the Yeong-Torquato algorithm, since a combination of these methods is used to reconstruct an SOFC anode based (i) on the entire FIB-SEM image and (ii) on only three image slices. Section 5 comprises a detailed evaluation of the geometrical and physical properties, as well as a thorough comparison between the original and the reconstructed microstructures. A conclusion and outlook are given in Section 6.

2 | Morphological Characterisation

In this section, crucial features of the microstructure that determine the effective physical properties are discussed. It is addressed which geometrical quantities are applied for the microstructure reconstruction in Section 4 via *MCRpy*, as certain descriptors are not differentiable and cannot be used in the differentiable optimisation procedure. In general, to characterise the microstructure from a statistical perspective, the voxel-based microstructure has to be mapped to a set of n_D stationary and translation-invariant descriptors $\mathbf{D}_i$. The characterisation function is introduced by

$$f_C : \mathbf{M} \mapsto \{\mathbf{D}_i\}_{i=1}^{n_D}. \quad (3)$$

In the case of SOFC electrodes, these statistical descriptors comprise the volume fractions (Section 2.1) of the individual components of the electrode up to the density of the reaction zones, the so-called double and triple phase boundaries (Section 2.5).

2.1 | Volume Fractions

The volume fraction of each constituent is essential to describe the microstructure statistically. For instance, an increasing Ni content in the anode leads to an improved electronic conductivity, but also to a larger thermal expansion coefficient mismatch in the entire fuel cell structure [39]. For computing the volume fractions, an indicator function $I^{(p)}$ for the respective phase p was introduced in Torquato [40] with

$$I^{(p)}(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{x} \text{ in phase } p, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

The indicator function can be utilised to compute the volume fraction of each phase

$$\varphi^{(p)} = \langle I^{(p)}(\mathbf{x}) \rangle \quad (5)$$

with the volume average defined by

$$\langle (\cdot) \rangle = \frac{1}{V} \int_B (\cdot) dV. \quad (6)$$

2.2 | Two-Point Functions

The morphology of the microstructure cannot be expressed by the volume fractions. Thus, correlation functions are used as statistical descriptors, namely the two-point correlation function and the lineal-path function. Mathematically, the auto correlation $S_2^{(p)}$ of phase p is expressed as

$$S_2^{(p)}(\mathbf{r}) = \langle I^{(p)}(\mathbf{x}) I^{(p)}(\mathbf{x} + \mathbf{r}) \rangle. \quad (7)$$

The vector $\mathbf{r}$, which has a specific length and orientation, can be imagined as line segments placed on a 3D image. Subsequently, the probability that the line segment starts and ends in the same respective phase p is calculated for each $\mathbf{r}$, see Section 5. In contrast to S_2 , the lineal-path function L calculates the probability that the entire line segment lies in a certain phase.

To save computational effort, especially in the reconstruction, *MCRpy* calculates S_2 and L only on 2D slices orthogonal to each Cartesian direction. Thus, for a 3D microstructure $\mathbf{M}^{I \times J \times K}$ with $I \times J \times K$ voxels, the two-point correlation function is applied $I + J + K$ times. Subsequently, S_2 and L of the slices of each direction are averaged over all planes. These averaged geometrical descriptors for each plane are included in the reconstruction process in Section 4. Furthermore,

so-called multigrid correlations are used to obtain a suitable and optimised descriptor for the differentiable framework. The main idea is to use the vectors $\mathbf{r}$ with a length up to a constant value and apply S_2 and L to the same microstructure but on different resolution representations of $\mathbf{M}$, so that both high-resolution, short-range correlations and low-resolution long-range correlations can be captured in a reasonable time. For more details, the interested reader is referred to Seibert et al. [36].

These correlation functions generally contain information about the form and the size of the features (e.g., the grains of the materials and the pores). If the line segment $|\mathbf{r}| = 0$, both S_2 and L contain the volume fraction of each phase. In addition, S_2 reflects the average distances between neighbouring regions, and provides information about the interface areas between the phases [40], but does not offer a high-fidelity description of the interfaces, such as the interfacial orientation. Furthermore, L incorporates additional information about the connectedness of the phases, which is important for achieving a suitable electronic and ionic conductivity. However, it is challenging to derive definitive figures from this characterisation [41].

2.3 | Percolation

The percolation $P^{(p)}$ refers to the formation of an interconnected and continuous network of conductive pathways due to the manufacturing process [42]. For the efficient transport of gas, the porous electrode needs to exhibit an interconnected pore phase to enable (i) the air and fuel to reach the reaction sites (double/triple phase boundary) and (ii) the reaction product (e.g., water in the anode) to escape. Porous inclusions that are not connected to the gas channels do not help in increasing the performance, as no oxidation process can occur in this inactive area. For an efficient conduction of electrons, the electronic conductor needs to be connected to the current collector, so that the electrons can be transported via the external circuit. In addition, the ceramic phase must form a continuous network to transport oxygen ions (i) from the cathode to the electrolyte and (ii) from the electrolyte to the reaction sites within the anode.

To calculate the percolation, each phase of the segmented 3D voxel dataset was grouped into continuous clusters using a cluster identification algorithm. Subsequently, the connection of each cluster to the necessary structural elements of the fuel cell is examined. The percolation probability of each phase $P^{(p)}$ can then be expressed as the fraction of voxels belonging to percolating clusters divided by the total voxels of that phase. If the representative volume element is too small, artificial truncation of clusters at the boundaries can prevent verification of connectivity, leading to large deviations from the true percolation characteristics. As the $\mathcal{RVE}$ size increases, the probability of capturing representative connectivity statistics improves, and boundary effects diminish. High percolation fractions are required for high electrical efficiencies, as they significantly affect the active double and TPB densities.

2.4 | Tortuosity

The tortuosity $\tau^{(p)}$ is a dimensionless crucial parameter for characterising the transport properties in porous fuel cell electrodes. However, there is generally not a unique definition of τ . The tortuosity can be classified into three groups according to the method of determination [43]: (i) the direct geometrical tortuosity which is determined by a 3D analysis of the underlying microstructure, (ii) the indirect physics-based τ which is extracted from effective properties that are identified by experiments or simulations, and (iii) the mixed tortuosity that depends on both a numerical simulation of a specific transport process and the path lengths of streamlines. In these three categories, there are plenty of different methods to determine the tortuosity [43, 44]. Since a pure geometrical parameter is desired that describes the morphology of interconnected paths, a direct geometrical approach is employed. Therefore, two different methods are implemented in Python: the so-called skeleton tortuosity and the geodesic tortuosity. The main goal of these methods is to find the effective path length L_{eff} from an initial voxel of a surface to the opposite surface of a 3D image with dimension L_0 , so that the basic definition of $\tau \geq 1$ can be applied:

$$\tau = \frac{L_{\text{eff}}}{L_0}. \quad (8)$$

For the skeleton tortuosity, the 3D microstructure needs to be segmented into its constituent phases, and for each segmented image, a medial axis skeleton is created [45]. The skeleton is transformed into a network representation that consists of vertices and nodes, that is, a 3D graph, which enables a cost-efficient computation. The shortest pathways from each voxel of a certain phase p , which lies on the skeleton on one face, to its facing surface are calculated using Dijkstra's algorithm [46]. The main difference of the geodesic tortuosity is the fact that no medial axis skeleton is produced for each

segmented image, and thus, all voxels of a certain phase and surface are considered as a starting point for searching the shortest-path lengths. This is computationally more demanding as the graph has significantly more considerable entries in Dijkstra's algorithm, especially for 3D images.

In total, $\tau^{(p)}$ can be calculated for each phase and all directions. As numerous interconnected paths can generally be determined, a distribution for the tortuosity results. Subsequently, statistical values such as the mean tortuosity $\bar{\tau}^{(p)}$, the median tortuosity $\tau_{\text{med}}^{(p)}$ or the standard deviation $\tau_{\text{std}}^{(p)}$ can be extracted for each phase and each direction. Often, only $\bar{\tau}^{(p)}$ is considered as the most important parameter.

2.5 | Double and Triple Phase Boundary

The electrochemical performance of electrodes strongly relies on the densities of the active parts of the triple phase boundary (ρ_{TPB}) in anodes and the double phase boundary (ρ_{DPB}) in cathodes [12, 47]. As already discussed, non-active parts of these reaction sites that are not percolated to the necessary layers in the fuel cell (see Section 2.3) do not contribute to the electrochemical performance. The TPB in Ni-YSZ anodes is the boundary where the phases of Ni, YSZ, and pores intersect, and the DPB in LSCF cathodes is simply the interface between the material and the pores. Lower densities ρ_{TPB} and ρ_{DPB} lead to higher polarisation resistances of the electrodes. However, the relationship between the densities of the reaction sites and the electrochemical performance is complex and nonlinear [22]. It is worth noting that the DPB density of Ni-YSZ anodes not only influences the performance, but also the durability and mechanical stability.

Even though additional geometrical descriptors are discussed in the literature, such as the particle and pore size distribution [21] or the constrictivity (which contains the quantitative analysis of bottlenecks) [48], this information is partly included in the two-point functions.

3 | Physical Properties

After geometrically characterising the electrode microstructure, the effective physical properties have to be determined to establish structure-property relationships. Therefore, the first-order computational homogenisation method [49, 50] is applied to extract the effective thermal, electric, and ionic conductivities (Section 3.1), the effective permeability (Section 3.2), and the effective mechanical parameters (Section 3.3). This method is based on the assumption of scale separation, that is, the microscopic length scale must be significantly smaller than the characteristic length governing macroscopic loading variations. In the following sections, constant macroscopic quantities are indicated by $(\cdot)$ and microscopic fluctuations are labelled with $\tilde{(\cdot)}$. For the homogenisation, the finite element method (FEM) with the open-source FE tool FEniCSx [51] is applied. FEniCSx utilises a range of preconditioners and solvers from PETSc [52], which can be used as desired. The homogenisation procedure was performed on the high-performance computing systems of the ZIH of TU Dresden. Thus, up to 100 CPUs were used for computationally demanding simulations such as the time-dependent creep behaviour. In the following, the equations to extract the non-mechanical quantities are only briefly discussed. A particular focus has been laid on mechanical parameters with a specific emphasis on the determination of effective creep parameters. Since the microstructural domain is large in order to capture all relevant statistical variations, iterative solvers are applied for all numerical simulations. The numerical details are briefly mentioned in Sections 3.1 to 3.3.

3.1 | Effective Conductivities

3.1.1 | Thermal Conductivity

To obtain the homogenised thermal behaviour, the steady-state heat equation $(9)_1$ with the well-known Fourier law $(9)_2$ is solved with the split of the primary variable into a constant part, a macroscopic gradient, and a fluctuation $(9)_3$.

$$\nabla \cdot \mathbf{Q} = 0, \quad \mathbf{Q} = -\kappa(T) \cdot \nabla T, \quad T = T_{\text{op}} + \overline{\nabla T} \cdot \mathbf{x} + \tilde{T}(\mathbf{x}) \quad (9)$$

Here, $\mathbf{Q}$ is the heat flux, T_{op} the operating temperature, κ the thermal conductivity, $\mathbf{x}$ the spatial coordinate and ∇ the nabla operator. To satisfy the equivalence of the thermal dissipation power density between the microscale and macroscale, that is, the Hill-Mandel lemma, three distinct boundary conditions can be applied [53]:

$$\tilde{T}(\mathbf{x}) = 0 \quad \forall \mathbf{x} \in \partial B, \quad \mathbf{Q}_n(\mathbf{x}) = \mathbf{Q}(\mathbf{x}) \cdot \mathbf{n}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B, \quad \tilde{T}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^+ = \tilde{T}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^-. \quad (10)$$

A uniform temperature gradient $(10)_1$, a uniform heat flux $(10)_2$ or periodic boundary conditions $(10)_3$ can be applied to fulfil the Hill-Mandel lemma. To determine the effective thermal conductivity κ^{eff} , at least three independent load cases must be calculated to extract the macroscopic quantity using the volume average (6):

$$\langle Q \rangle = -\kappa^{\text{eff}} \cdot \langle \nabla T \rangle. \quad (11)$$

3.1.2 | Electric Conductivity

The continuity equation $(12)_1$ with Ohm's law $(12)_2$ is solved by FEM to extract the effective electronic conductivity $\lambda^{e-, \text{eff}}$. The driving force of the flow of electrons in the electric conductor (e.g., Ni), that leads to the current density J^{e-} , is the macroscopic electric field $\bar{E}$. Therefore, the electric scalar potential ϕ can be split as shown in Equation $(12)_3$.

$$\nabla \cdot J^{e-} = 0, \quad J^{e-} = \lambda^{e-}(T) \cdot E, \quad \phi = -\bar{E} \cdot x + \tilde{\phi}(x) \quad (12)$$

The boundary conditions can be analogously derived as for the thermal field to agree with the equivalence of the ponderomotive power densities:

$$\tilde{\phi}(x) = 0 \quad \forall x \in \partial B, \quad J_n^{e-}(x) = J^{e-}(x) \cdot n(x) \quad \forall x \in \partial B, \quad \tilde{\phi}(x) \quad \forall x \in \partial B^+ = \tilde{\phi}(x) \quad \forall x \in \partial B^-. \quad (13)$$

For the macroscopic constitutive behaviour, Ohm's law is employed again as

$$\langle J^{e-} \rangle = \lambda^{e-, \text{eff}} \cdot \langle E \rangle \quad \text{with} \quad \langle E \rangle = \langle -\nabla \phi \rangle. \quad (14)$$

From a numerical perspective, the matrix is diagonally scaled with the help of the Jacobi preconditioner. This is followed by the application of the conjugate gradient method to solve the system of equations iteratively. The same numerical settings were applied for determining the other conductivities.

3.1.3 | Ionic Conductivity

The gradient of the electrochemical potential ω is the driving force for the flow of ions in ion-conducting materials, such as YSZ in SOFC anodes. Thus, the linear material law $(15)_2$ is used for the continuity of ions $(15)_1$ with a split of ω in Equation $(15)_3$.

$$\nabla \cdot J^{O^{2-}} = 0, \quad J^{O^{2-}} = -\frac{\lambda^{O^{2-}}(T)}{z^{O^{2-}} F} \cdot \nabla \omega, \quad \omega = \bar{\nabla} \omega \cdot x + \tilde{\omega}(x). \quad (15)$$

The ionic current density $J^{O^{2-}}$ is also related to the valency of oxygen ions $z^{O^{2-}}$ and the Faraday constant F . The three typical types of boundary conditions

$$\tilde{\omega}(x) = 0 \quad \forall x \in \partial B, \quad J_n^{O^{2-}}(x) = J^{O^{2-}}(x) \cdot n(x) \quad \forall x \in \partial B, \quad \tilde{\omega}(x) \quad \forall x \in \partial B^+ = \tilde{\omega}(x) \quad \forall x \in \partial B^-. \quad (16)$$

can be set. The averaged quantities of the gradient of the electrochemical potential and the flow of oxygen ions are used to calculate the effective ionic conductivity $\lambda^{O^{2-}, \text{eff}}$ using

$$\langle J^{O^{2-}} \rangle = -\frac{\lambda^{O^{2-}, \text{eff}}(T)}{z^{O^{2-}} F} \langle \nabla \omega \rangle. \quad (17)$$

3.2 | Effective Permeability

Due to very small fluid velocities v (and thus small Reynolds numbers) and small gradients of the pressure p , the Stokes equations with the mass balance $(18)_1$ and the balance of linear momentum $(18)_2$ can be used to describe the flow of fuel and air within electrodes at the microscale. The constitutive behaviour $(18)_3$ of the fluid can be characterised by assuming a Newtonian fluid with the dynamic viscosity μ . The pressure is again split into a macroscopic and fluctuation part, see Equation $(18)_4$.

$$\nabla \cdot v = 0, \quad \nabla \cdot \sigma^f = 0, \quad \sigma^f = -pI + 2\mu(T) d, \quad p = \bar{\nabla} p \cdot x + \tilde{p}(x). \quad (18)$$

Here, σ^f is the Cauchy stress tensor of the fluid, $\mathbf{I}$ the unit tensor, and $\mathbf{d}$ the strain rate tensor. From a numerical point of view, it is essential to precondition the “stiffness matrix” to ensure convergence of the resulting saddle point problem. Thus, the form of the preconditioner is $\mathbf{P} = \int_B \nabla \mathbf{v} : \nabla \delta \mathbf{v} \, dV + \int_B \tilde{p} \, \delta \tilde{p} \, dV$. The preconditioned system then acts as an approximation operator within the Krylov iteration. As an iterative solver, the generalised minimal residual (GMRES) method is employed, which is particularly suited for nonsymmetric saddle point problems. Convergence is required when the relative residual is reduced by at least five orders of magnitude (10^{-5}). For the inversion of $\mathbf{P}$, the HyPre library with the algebraic multigrid method BoomerAMG is used, which provides an efficient and scalable preconditioning strategy. To satisfy the Hill-Mandel lemma, only two types of boundary conditions can be prescribed [54], such that (i) the pressure fluctuations are set to zero at all boundaries (19)₁ and the normal component of the velocity is zero if the macroscopic pressure gradient $\mathbf{v}$ is zero in the direction of the normal vector (19)₂; or (ii) the pressure fluctuations and all components of the velocity fulfil periodicity (20). In addition, all velocities and fluid pressures are set to zero within the material phases B^{mat} (21), as no fluid flows through the particles, but only through the porous phase. Moreover, no-slip conditions are selected so that the velocity at the interface is zero (22).

$$\tilde{p}(\mathbf{x}) = 0 \quad \forall \mathbf{x} \in \partial B, \quad v_n(\mathbf{x}) = \mathbf{v}(\mathbf{x}) \cdot \mathbf{n}(\mathbf{x}) = 0 \quad \forall \mathbf{x} \in \partial B \text{ if } \overline{\nabla p} \cdot \mathbf{n} = 0, \quad (19)$$

$$\tilde{p}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^+ = \tilde{p}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^-, \quad \mathbf{v}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^+ = \mathbf{v}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^-, \quad (20)$$

$$\tilde{p}(\mathbf{x}) = 0 \quad \forall \mathbf{x} \in B^{\text{mat}}, \quad \mathbf{v}(\mathbf{x}) = \mathbf{0} \quad \forall \mathbf{x} \in B^{\text{mat}}, \quad (21)$$

$$\mathbf{v}(\mathbf{x}) = \mathbf{0} \quad \forall \mathbf{x} \in \partial \mathbf{I}. \quad (22)$$

On the macroscale, Darcy’s law (23) is commonly used to describe the flow of fluid in porous media with an effective permeability $\mathbf{K}^{\text{eff}}$ as

$$\langle \mathbf{v} \rangle = \frac{\mathbf{K}^{\text{eff}}}{\mu} \langle \nabla p \rangle. \quad (23)$$

3.3 | Effective Mechanical Parameters

In this section, the homogenisation to analyse the effective elasticity tensor $\mathbb{C}^{\text{eff}}$, the thermal expansion α^{eff} as well as the creep parameters A^{eff} and n^{eff} is explained in detail. The stresses that arise in fuel cell electrodes need to fulfil the balance of linear momentum (24)₁. Linear elastic constitutive behaviour (24)₂ is assumed for the Cauchy stress σ . The total strain ϵ is assumed to be additively decomposed into an elastic, a thermal and a creep term, see Equation (24)₃. The split of the primary variable $\mathbf{u}$ can be seen in Equation (24)₄.

$$\nabla \cdot \sigma = \mathbf{0}, \quad \sigma = \mathbb{C}(T) : \epsilon^{\text{el}}, \quad \epsilon^{\text{el}} = \epsilon - \epsilon^{\text{th}} - \epsilon^{\text{cr}}, \quad \mathbf{u} = \bar{\epsilon} \cdot \mathbf{x} + \tilde{\mathbf{u}}(\mathbf{x}). \quad (24)$$

Here, the symmetric total strain tensor can be mathematically expressed for the assumption of infinitesimal deformations as $\epsilon = \frac{1}{2}(\nabla \mathbf{u} + \mathbf{u} \nabla)$. Due to changes in the temperature T , thermal strains occur with $\epsilon^{\text{th}} = \alpha(T)(T - T_{\text{ref}})$ with T_{ref} as the reference temperature in a stress-free state. The stress-free state is related to the sintering process of the electrodes. The chemical expansion [55] due to changes in the oxygen vacancies—which is considered, for instance, in Muramatsu et al. [56] for the scale transition—is expressed as $\epsilon^{\text{chem}} = \beta ([V_{\text{O}}^{\bullet}] - [V_{\text{O}}^{\bullet}]_{\text{ref}}) \mathbf{I}$ with the chemical expansion coefficient β and the oxygen stoichiometry $V_{\text{O}}^{\bullet}$ in the actual and reference state. Please note that the effective chemical strain is not computed in this work. Residual stresses σ_0 arising during cooling after sintering at high temperature and resulting from volume changes during NiO reduction should be considered at the macroscale, as they can influence mechanical behaviour and microstructural stability. However, since our model does not include microstructural evolution, these stresses are not expected to affect the predicted effective mechanical properties.

Analogously to the conductivities, three different boundary conditions can be prescribed to fulfil the equivalence of the macroscopic and averaged microscopic rates of internal mechanical work with $\langle \sigma : \dot{\epsilon} \rangle = \bar{\sigma} : \dot{\bar{\epsilon}}$. Therefore, either linear displacement boundary conditions (25)₁, constant tractions (25)₂ or periodic boundary conditions with periodic displacement fluctuations (25)₃ and antiperiodic tractions (25)₄ can be prescribed:

$$\begin{aligned} \tilde{\mathbf{u}}(\mathbf{x}) = \mathbf{0} \quad \forall \mathbf{x} \in \partial B, \quad \bar{\mathbf{t}}(\mathbf{x}) = \bar{\sigma}^T(\mathbf{x}) \cdot \mathbf{n}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B, \quad \tilde{\mathbf{u}}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^+ = \tilde{\mathbf{u}}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^-, \\ \mathbf{t}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^+ = -\mathbf{t}(\mathbf{x}) \quad \forall \mathbf{x} \in \partial B^-. \end{aligned} \quad (25)$$

The effective parameters for each strain contribution can be computed separately. Thus, to compute the macroscopic elastic properties, only elastic strains are considered with $\epsilon^{\text{th}} = \epsilon^{\text{cr}} = \mathbf{0}$. The GMRES iterative solver is utilised in conjunction with a multigrid preconditioner from the HyPre library to solve the mechanical field. Specifically, BoomerAMG is used with 15 smoothing levels and a strong connection threshold of 0.7 to provide an efficient, scalable preconditioning strategy. An absolute convergence criterion of 10^{-8} is set. The effective stiffness tensor in the purely elastic range can be computed by evaluating at least six load cases with linearly independent macroscopic strains $\bar{\epsilon}$. Subsequently, the classical averaging can be applied with $\langle \sigma \rangle = \mathbb{C}^{\text{eff}} : \langle \epsilon \rangle$. However, to automatically obtain a symmetric elasticity tensor, the global form of the balance of linear momentum (equation (26)) is employed. Here, the Betti reciprocity theorem [57] with $\sigma^1 : \epsilon^2 = \sigma^2 : \epsilon^1$ is utilised, which states that the work done by the stress tensor σ^1 on the strain state ϵ^2 is equal to the work done by the stress tensor σ^2 on the strain state ϵ^1 . Thus, the macroscopic mechanical homogeneity condition is expressed as

$$\langle \mathbb{C}(\mathbf{x}) : \epsilon^m(\mathbf{u}^m(\mathbf{x})) : \epsilon^n(\mathbf{u}^n(\mathbf{x})) \rangle = \mathbb{C}^{\text{eff}} : \langle \epsilon^m \rangle : \langle \epsilon^n \rangle, \quad (26)$$

with m and n being the number of load cases. In total, 21 independent combinations can be calculated in the post-processing so that 21 independent constants of $\mathbb{C}^{\text{eff}}$ can be calculated. In general, the effective elasticity tensor has no symmetry planes due to discretisation errors of the FE method and is completely anisotropic. However, to (i) neglect discretisation errors, (ii) gain more precise information about the mechanical behaviour of SOFC electrodes, and (iii) be able to express the macroscopic properties as a strain energy, which can be used to be implemented in macroscopic simulations, the closest elastic tensor of the triclinic symmetry to a material class of lower symmetry is optimised by the minimisation of the loss function (27)₁. This results in the detection of an effective elasticity tensor $\mathbb{C}^{\text{sym}}$ belonging to a specific symmetry group $\mathcal{G}$. The Frobenius norm (27)₂ can be interpreted as the distance between the anisotropic $\mathbb{C}^{\text{eff}}$ and $\mathbb{C}^{\text{sym}}$.

$$\mathbb{C}^{\text{sym}} = \arg \min_{\mathbb{C}} \mathcal{L}(\mathbb{C}(\mathcal{G}), \mathbb{C}^{\text{eff}}), \quad \mathcal{L}(\mathcal{G}) = \|\mathbb{C} - \mathbb{C}^{\text{eff}}\|_F. \quad (27)$$

The effective thermal expansion α^{eff} can be obtained by isolating the other contributions of the elastic strains ($\epsilon = \epsilon^{\text{cr}} = \mathbf{0}$) and by prescribing a specific temperature change from the reference temperature. Thus, all six components of the symmetric second-order tensor can be computed by one load case with

$$\langle \sigma \rangle = -\mathbb{C}^{\text{eff}} : \alpha^{\text{eff}}(T - T_{\text{ref}}). \quad (28)$$

SOFCs are typically operated at high temperatures between 600°C and 1000°C. An inelastic mechanical process that needs to be taken into account is creep, as this mechanism occurs above 50% of the melting temperature (e.g. $T_{\text{m}}^{\text{Ni}} = 1450^\circ\text{C}$ for nickel). In the literature, creep is already addressed as a critical process that affects the durability of fuel cells, but the determination of effective creep parameters is often neglected. In general, creep is divided into three stages: (i) the primary stage with a decreasing creep rate over time, (ii) the second steady stage with a constant creep rate, and (iii) the tertiary creep with an increasing creep rate until the component is completely damaged [58]. The internal variable, the creep strain rate $\dot{\epsilon}^{\text{cr}}$ can be derived from the creep potential Q by Equation (29)₁. For the long-term operation of fuel cells, it is assumed that the secondary stage (ii) is the most significant creep process in the service lifetime of SOFCs. Therefore, the classical Norton law (29)₂ [59] is chosen as it can accurately describe this constant rate stage for metals and alloys, and it is also employed to calibrate the creep parameters A and n of Ni and YSZ [60]. The equivalent stress σ_{eq} is calculated by Equation (29)₃ with the fourth-order tensor $\mathbb{M}$ that can generally express anisotropy. The evolution of the internal variable $\dot{\epsilon}^{\text{cr}}$ results in Equation (29)₄.

$$\dot{\epsilon}^{\text{cr}} = \frac{\partial Q}{\partial \sigma}, \quad Q = \frac{A}{n+1} \sigma_{\text{eq}}^{n+1}, \quad \sigma_{\text{eq}} = \sqrt{\sigma : \mathbb{M} : \sigma}, \quad \dot{\epsilon}^{\text{cr}} = A \sigma_{\text{eq}}^{n-1} \mathbb{M} : \sigma. \quad (29)$$

In the following, the numerical details for implementing this model in the software environment FEniCSx [51] are explained. A predictor-corrector method is employed to compute the creep strain in every iteration. Therefore, the incremental change of stress for each time increment due to an external load can be expressed as

$$\Delta \sigma = \underbrace{\mathbb{C} : \Delta \epsilon}_{\text{Predictor}} - \underbrace{\mathbb{C} : \Delta \epsilon^{\text{cr}}}_{\text{Corrector}}. \quad (30)$$

First, the trial stress is computed by the predictor step with $\sigma_{\text{trial}} = \mathbb{C} : \Delta \epsilon$ due to an actual traction vector $\mathbf{t}$. Therefore, the global system of equations is solved at all Gauss points in Voigt notation:

$$\underline{\underline{K}}_{\text{tang}} \underline{\Delta u} = -R, \quad \underline{\underline{K}}_{\text{tang}} \underline{\Delta u} := \int_B \underline{\epsilon}(\mathbf{u}) \cdot \left(\underline{\underline{C}}^{\text{algo}} \cdot \underline{\epsilon}(\mathbf{v}) \right) dV, \quad R := \int_B \underline{\epsilon}(\mathbf{v}) \cdot \underline{\sigma} dV - \int_{\partial B} \underline{\mathbf{t}n} \cdot \mathbf{v} dA. \quad (31)$$

All quantities in Voigt notation are underlined to express vectors $\underline{(\cdot)}$ and matrices $\underline{(\cdot)}$, respectively. Here, $\underline{K}_{\text{tang}}$ is the stiffness tangent in matrix notation, $\underline{\Delta u}$ the incremental displacement and $\underline{R}$ the residuum. The left-hand side can be assembled by the strain expression with the displacement $\underline{u}$ and the test function $\underline{v}$, as well as the algorithmically consistent tangent $\underline{C}^{\text{algo}}$. For the right-hand side, the stress $\underline{\sigma}$ is additionally needed to the traction vector to capture the residuum. Both sides are integrated over the entire domain $\mathcal{B}$. After calculating the trial stress $\underline{\sigma}_{\text{trial}}$, the deviatoric part of the trial stress $\underline{S}_{\text{trial}}$ is required for calculating the incremental creep strain. The deviatoric part of a tensor is generally calculated by $\underline{S} = \underline{\sigma} - 1/3 \text{tr}(\underline{\sigma}) \underline{I}$, with $\text{tr}(\cdot)$ being the trace of a second-order tensor and $\underline{I}$ being the unity tensor. The creep strain increment can be computed at every Gauss point by Equation (32)₁ and depends on a scalar incremental damage Δp and the tensor that determines the direction of creep $\underline{a}$. These quantities are computed within the corrector step with Equations (32)₂ and (32)₃. For extensions of the model, such as plasticity and different time discretisation, the reader is referred to Refs. [61, 62].

$$\Delta \underline{\epsilon}^{\text{cr}} = \underline{a} \Delta p, \quad \underline{a} = \sqrt{\frac{3}{2}} \frac{\underline{S}_{\text{trial}}}{\|\underline{S}_{\text{trial}}\|}, \quad \Delta p = \frac{\underline{a} : \underline{C} : \underline{\Delta \epsilon} + \frac{\sigma_{\text{eq}}}{\theta n}}{\underline{a} : \underline{C} : \underline{a} + \frac{1}{\Delta t \theta A n \sigma_{\text{eq}}^{n-1}}}. \quad (32)$$

It is worth noting that the implicit time-stepping scheme with $\theta = 1$ is implemented. The algorithmic consistent tangent is used in the assembling of the left-hand side (31)₂ for better convergence rates. It can be analytically derived by

$$\underline{C}^{\text{algo}} := \frac{d\underline{\sigma}}{d\underline{\epsilon}} = \underline{C} - \frac{\underline{C} : \underline{a} \otimes \underline{a} : \underline{C}}{\underline{a} : \underline{C} : \underline{a} + \frac{1}{\Delta t \theta A n \sigma_{\text{eq}}^{n-1}}}. \quad (33)$$

After reaching a relative displacement tolerance lower than a prescribed one, a new time step is calculated. The procedure of the predictor-corrector method is summarised in Algorithm 1.

ALGORITHM 1 | Predictor-corrector algorithm of the Norton model.

```

Initialize  $t = 0$ ,  $\Delta t = 0.01$  s,  $\underline{C}^{\text{algo}} := \underline{C}$ ,  $\underline{t} = 0$ 
while  $t \leq t_{\text{final}}$  do
     $t \leftarrow t + \Delta t$ 
    Update traction vector  $\underline{t}$ 
    while  $\mathcal{R} > \text{tol}$  and  $n_{\text{it}} < n_{\text{it,max}}$  do
        Predictor step
        Assembling of the residuum
        Assembling of the stiffness matrix
        Solve the global system of equations:  $\underline{\Delta u}$ 
        Update total displacement  $\underline{u} \leftarrow \underline{u} + \underline{\Delta u}$ 
        Computation of total strain increment and trial stress  $\underline{\Delta \epsilon}$ ,  $\underline{\sigma}_{\text{trial}}$ 
        Corrector step
        Determination of the creep strain increment  $\underline{\Delta \epsilon}^{\text{cr}}$ 
        Determination of the true stress  $\underline{\sigma}$ 
        Determination of the algorithmic consistent tangent  $\underline{C}^{\text{algo}}$ 
        Determination of new residuum
        Relative displacement residual  $\mathcal{R} = \frac{\|\underline{\Delta u}\|_2}{\|\underline{u}\|_2}$ 
         $n_{\text{it}} \leftarrow n_{\text{it}} + 1$ 
    end while
    Update creep strain  $\underline{\epsilon}^{\text{cr}} \leftarrow \underline{\epsilon}^{\text{cr}} + \underline{\Delta \epsilon}^{\text{cr}}$ 
    Update total strain  $\underline{\epsilon} \leftarrow \underline{\epsilon} + \underline{\Delta \epsilon}$ 
    Update stress  $\underline{\sigma}$ 
    Compute creep strain rate  $\dot{\underline{\epsilon}}^{\text{cr}}$ 
    Update  $\Delta t$  according to adaptive time step method
end while

```

Several load cases m are performed in which external loads are applied to the $\mathcal{RVE}$ in different directions and planes with varying magnitudes. For each load case, symmetry boundary conditions for the displacement are employed due to better convergence rates compared to the use of constant tractions at the opposite surfaces, and preventing rigid body movements only at a certain degree of freedom. After the convergence of the creep strain rate $\dot{\epsilon}^{\text{cr}}$ to a constant value, the stress σ and the creep strain rate $\dot{\epsilon}^{\text{cr}}$ are volume averaged by Equation (6). To extract the effective properties that are generally anisotropic, the Norton law (29)₄ is minimised to obtain A^{eff} , n^{eff} and $\mathbb{M}^{\text{eff}}(\mathcal{G})$, see Equation (6). The tensor $\mathbb{M}^{\text{eff}}$ can mathematically describe the anisotropy of creep and is assumed to have the same symmetry group $\mathcal{G}$ as $\mathbb{C}^{\text{sym}}$. In the following, this parameter set is summarised as $\mathcal{P}^{\text{cr}}$. Moreover, the Frobenius norm is used to determine the effective creep parameters [63]:

$$\mathcal{P}^{\text{cr}} = \arg \min_{\mathcal{P}^{\text{cr}}} \sum_m \|A^{\text{eff}} \sigma_{\text{eq}}^{n^{\text{eff}}-1}(\langle \sigma_m \rangle, \mathbb{M}^{\text{eff}}) \mathbb{M}^{\text{eff}} : \langle \sigma_m \rangle - \langle \dot{\epsilon}_m^{\text{cr}} \rangle\|_F. \quad (34)$$

4 | Microstructure Reconstruction

The reconstruction of the microstructure is separated into two parts: First, the reconstruction is executed using differentiable microstructure characterisation and reconstruction (DMCR) to efficiently generate a morphologically plausible solution [36]. Second, an adapted version of the well-known Yeong-Torquato algorithm [64] is used for improving this solution by reducing the noise and fine-tuning certain specific descriptors. The general idea of such a two-stage approach is presented and validated in [37]; however, the relevance of connectivity and phase boundaries in fuel cell applications motivates some modifications. After a brief introduction, these modifications are outlined in the following.

Both methods, DMCR and the Yeong-Torquato algorithm-type post-processing, are based on the optimisation problem

$$\mathbf{M}^{\text{rec}} = \arg \min_{\mathbf{M}} \mathcal{L}(\{\mathbf{D}_i(\mathbf{M}), \mathbf{D}_i^{\text{des}}\}_{i=1}^{n_D}). \quad (35)$$

to generate reconstructed microstructures $\mathbf{M}^{\text{rec}}$ from desired descriptors $\mathbf{D}_i^{\text{des}}$ with $1 \leq i \leq n_D$, where n_D denotes the number of descriptors. The loss function $\mathcal{L}$ is the error measure and is defined as the weighted sum of mean squared errors (MSEs) between the descriptors $\mathbf{D}_i$ and $\mathbf{D}_i^{\text{des}}$ that results from the characterisation function (3):

$$\mathcal{L}(\mathbf{M}) = \sum_{i=1}^{n_D} \|\mathbf{D}_i - \mathbf{D}_i^{\text{des}}\|_{\text{MSE}}. \quad (36)$$

In DMCR, Equation (35) is solved by gradient-based optimisation algorithms. Hence, the derivative $d\mathcal{L}/d\mathbf{M}$ needs to be computed for the update of $\mathbf{M}^{\text{rec}}$ in every iteration. Consequently, the descriptors must be continuously differentiable. This restriction is accepted in view of the very high efficiency of gradient-based optimisation algorithms. Although these can only identify local minima, experience shows that with the right choice of the descriptors, these local minima are often acceptable, except for some noise that needs to be removed. Contrary, the Yeong-Torquato algorithm solves the optimisation problem (35) by swapping pairs of voxels using the simulated annealing optimiser. It has the potential of finding global optima, but in a practical 3D reconstruction of a complex, anisotropic structure, hyperparameters that enable the global optimum to be found with certainty are computationally unfeasible. Extensive works in the literature address this issue by introducing modifications such as different-phase neighbour sampling rules [65], efficient descriptor updates [66, 67], or the restriction to radial, that is, isotropic, descriptors [67]. An overview is given in Bostanabad et al. [68], although the effect of such modifications on the resulting structures is not always clear and requires further investigations [69]. This motivates a two-step approach, where the DMCR solution serves as an initialisation to a post-processing stage based on the Yeong-Torquato algorithm, combining the advantages of both methods [37].

With the specific microstructure and challenges of solid oxide fuel cell electrodes in mind, the approach from Seibert et al. [37] is adapted in three ways, namely (i) the choice of descriptors, (ii) the extension to truly 3-dimensional descriptors, and (iii) the choice of pixel selection probability. These are laid out in the following.

4.1 | Descriptor Choice

As descriptors $\mathbf{D}_i$, the volume fractions $\varphi^{(p)}$, the two-point correlation functions $S_2^{(p)}$ and the densities of the triple and double phase boundaries $\rho_{\text{TPB}}/\rho_{\text{DPB}}$ are applied. Since the lineal-path function $L^{(p)}$ is generally not differentiable, an approximation $\tilde{L}^{(p)}$ is derived to be applicable via the MCRpy tool [38]. Even though the percolation P and the tortuosity

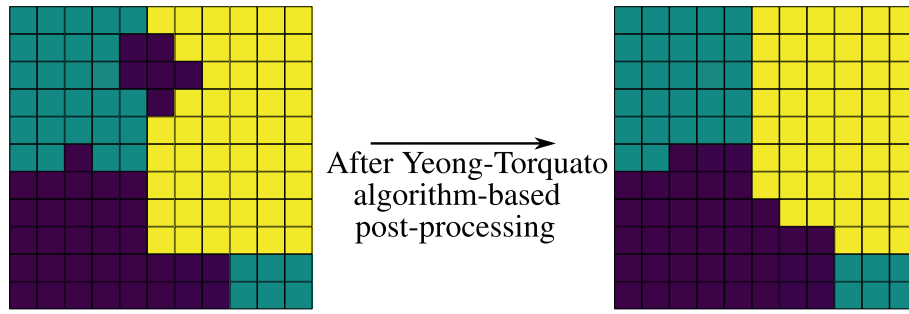


FIGURE 2 | Visualisation of the mutation rule: Two pixels in the porous composite are selected to remove non-percolated clusters (e.g., the purple island) and reduce ρ_{TPB} simultaneously. Please note that swapping two pixels is only accepted if the total loss (36) is reduced after the iteration.

τ are still important geometrical quantities, they are not directly used during the reconstruction process. The main reason is that both features are not differentiable. Moreover, they are also not employed by the Yeong-Torquato algorithm in the second step of the reconstruction, as this algorithm requires cheap descriptors, that is, descriptors which can be computed very quickly in order to be efficient. The computation of the mean tortuosity $\bar{\tau}^{(p)}$, however, is very expensive, especially for large 3D microstructural images. The tortuosity can also not be computed in each 2D slice, as the percolation paths are not entirely in the 2D plane. Thus, the mean tortuosity and the percolation of each phase between the original image and the reconstruction are only compared afterwards.

4.2 | Extension to 3D

In DMCR, the computation and automatic differentiation of 3D descriptors take unfeasible amounts of memory. Therefore, descriptors are only computed and compared to the reference on 2D slices, where a loop iterates through all 2D slices of the structure in each of the three spatial directions. In *MCRpy*, the same applies to the Yeong-Torquato algorithm, whereby an increased efficiency is reached by computing descriptors in each iteration only on slices that contain modified pixels. However, both the double phase boundary and the triple phase boundary are inherently three-dimensional quantities that are difficult to reduce to 2D. Therefore, *MCRpy* is extended to handle truly 3-dimensional descriptors, and 3D implementations of ρ_{TPB} , ρ_{DPB} and $S_2^{(p)}$ are developed. These are used in the Yeong-Torquato algorithm-based post-processing.

4.3 | Pixel Selection Probability

In the Yeong-Torquato algorithm-based post-processing, two randomly selected pixels are swapped in each iteration to see if this reduces the loss function (36). However, as the solution becomes closer to the optimum, this is increasingly unlikely, drastically reducing the convergence speed. This is especially problematic in the present work, where a post-processing has to be carried out. Therefore, pixels are not sampled naively, but based on a user-defined probability distribution. For example, it is common to sample only pixels at phase interfaces [65, 68], or more generally, to make a pixel's selection probability dependent on its number of different-phase neighbours. With solid oxide fuel cell electrodes, it is clear that material islands surrounded by air or fuel are impossible; however, it is difficult to directly enforce this criterion in DMCR. At the same time, noise tends to over-estimate ρ_{TPB} and ρ_{DPB} . For this reason, a hybrid pixel selection criterion is chosen, where one pixel is chosen from an island's phase boundary and the other is selected based on its number of different-phase neighbours, see Figure 2.

5 | Results

In this section, both geometrical descriptors and effective physical properties are analysed and compared between the original and reconstructed microstructures. An anode consisting of Ni and YSZ was obtained by Holzer et al. [70] using the FIB-SEM methodology. The original microstructure $\mathbf{M}^{\text{orig}}$ (see Figure 3 (left)) has fine grains, and the image was taken before redox reactions. A representative part consisting of 600 voxels in each direction was downsampled by the factor of 1/3 to decrease the resolution and thus, the computational costs. Therefore, an image with 200^3 voxels—corresponding to

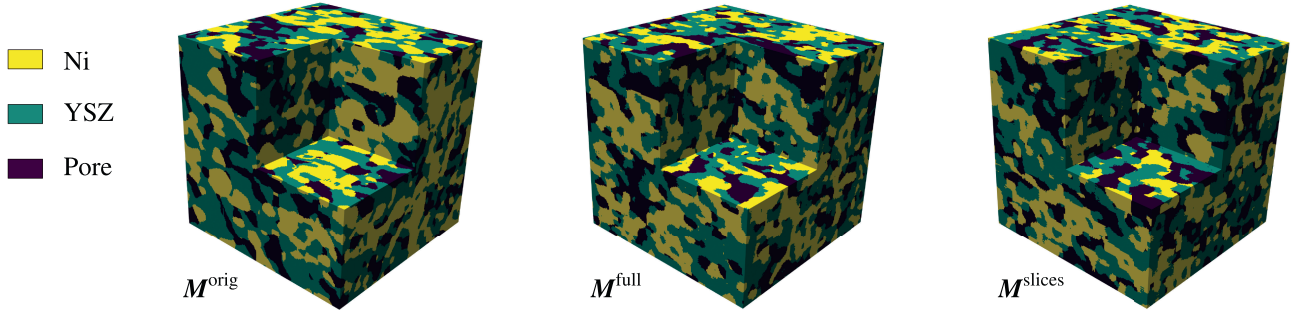


FIGURE 3 | Microstructures of the porous solid oxide fuel cell anode: original microstructure from FIB-SEM tomography [70] M^{orig} (left), the reconstruction based on the full 3D microstructure M^{full} (centre) and the reconstruction based on three experimental orthogonal slices of the image M^{slices} (right).

a physical volume of $(12\mu\text{m})^3$ —was geometrically and physically characterised. Two microstructures were reconstructed based on (i) the whole 3D original microstructure, that is, all slices were considered, and (ii) only three orthogonal slices. These microstructures (M^{orig} , M^{full} , M^{slices}) are shown in Figure 3.

5.1 | Morphological Descriptors

The main characteristic features that describe the anode microstructure of SOFCs (see Section 2) are evaluated for the original image and compared to the reconstructions. In Figure 4, the morphological descriptors of the three microstructures are summarised. In addition, each morphological descriptor is analysed in detail in the following.

The anode consists of a Ni–YSZ composite with 32.55 % nickel, 41.63 % yttrium-stabilised zirconia and 25.81 % pores. The reconstructions have only marginal differences in the *volume fractions*, see Figure 4a. Most of the continuous network of each phase is connected to the necessary structured elements, since the *percolation* of each phase is close to 1. Although the percolation is not used in the reconstruction process as a descriptor, the percolation of each phase of both reconstructions reveals a nearly perfect agreement with the original microstructure.

Exemplary corresponding heat maps of the *two-point correlation function* $S_{2,z}^{\text{pore}}$ and the *lineal-path function* L_z^{pore} are shown for M^{orig} in Figure 5 to give a graphical impression of these statistical functions. The brighter the colour, the higher the probability that the line segment with a certain length and orientation starts and ends within the pore phase or lies entirely in the pore phase, respectively. Since the multigrid procedure is used, only short-range correlations with a high resolution are shown. The lines at L_z^{pore} in the 0, 45, 90, 135, etc. degree direction are the result of discretising $\mathbf{r}$ as a line when calculating $L^{(p)}$. For simplicity, the Bresenham algorithm is used in this context, which is not capable of anti-aliasing. The use of alternative line algorithms could be a solution. However, this would require other computational steps to be adapted, which would increase the complexity.

In Figure 4c, the morphological MSEs $\mathcal{E}^{\text{mor}}$ with

$$\mathcal{E}^{\text{mor}} = \|(\cdot)_{\text{rec}} - (\cdot)_{\text{orig}}\|_{\text{MSE}} / \|(\cdot)_{\text{orig}}\|_{\text{MSE}} \quad (37)$$

of the two-point correlation function and the lineal-path function between M^{orig} and $\{M^{\text{full}}, M^{\text{slices}}\}$ are depicted, as it is hard to derive a quantitative number from these functions. It is worth noting that the reconstructed 3D microstructure M^{slices} is also compared to the original microstructure M^{orig} , even though it only contains the information of three orthogonal slices and not of the full 3D image. Due to that reason, the relative errors of the correlation functions are naturally lower for M^{full} than for M^{slices} . However, the relative errors for M^{slices} are still in good agreement with M^{orig} .

Estimating the *TPB density* of voxelised images is a challenge. In this work, the length l_{TPB} is determined by simply using the edge lengths of the voxels. Although this leads to an overestimation of l_{TPB} due to the voxelised discretisation, a good comparability is achieved between the microstructures. Further methods for calculating this length can be found in Shikazono et al. [71] The density of the electrochemically active TPB is higher for the reconstructions compared to the original microstructure: $\rho_{\text{TPB}}(M^{\text{orig}}) = 3.54\mu\text{m}^{-2}$, $\rho_{\text{TPB}}(M^{\text{full}}) = 4.48\mu\text{m}^{-2}$, $\rho_{\text{TPB}}(M^{\text{slices}}) = 3.99\mu\text{m}^{-2}$. Local minima

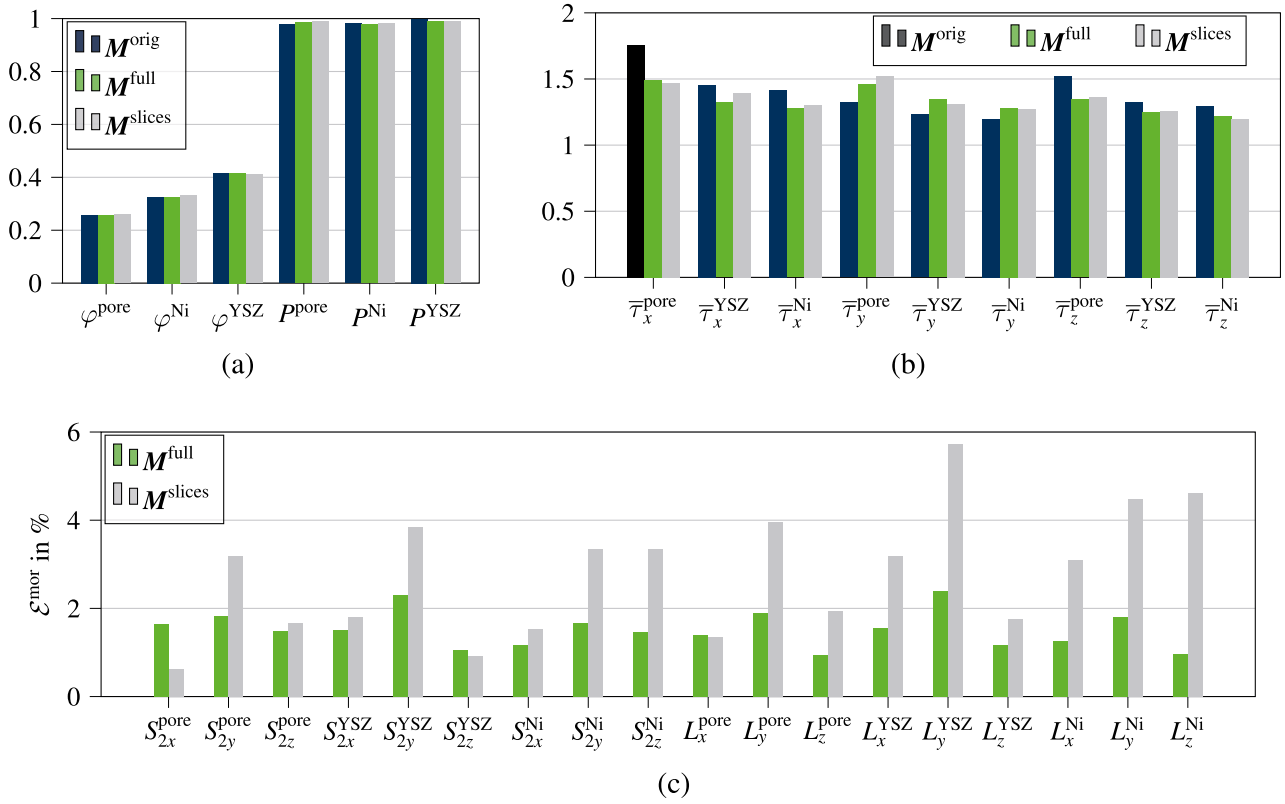


FIGURE 4 | Comparison of morphological descriptors between the original and the reconstructed microstructures. (a) Volume fractions $\varphi^{(p)}$ and percolations $P^{(p)}$ (b) Mean tortuosities $\bar{\tau}_i^{(p)}$ (c) Relative errors $\mathcal{E}^{\text{mor}}$ for two-point correlation function $S_{2i}^{(p)}$ and lineal-path function $L_i^{(p)}$ between original and reconstructed microstructures.

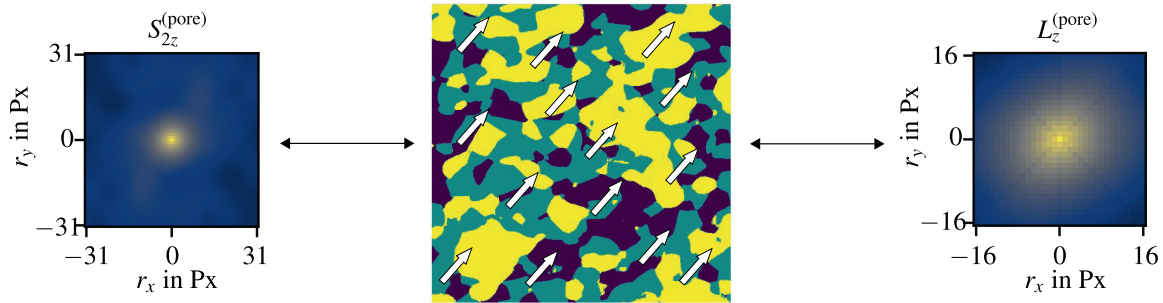


FIGURE 5 | Evaluation of the two-point correlation function (left) and the lineal-path function (right) for the pore phase and at the x - y plane for short-range correlations. The line segments are exemplarily placed into a 2D slice of the microstructure (centre) for visualising the procedure in order to determine the respective probabilities.

in the first step of the optimisation procedure (DMCR) are the reason that it does not converge to the desired TPB density. Even though the Yeong-Torquato algorithm-based post-processing reduces the length of the TPB by lowering noise in the second step, 45,000 iterations are not enough to decrease it to the desired density. This does not affect the results in Section 5.2, since the model does not consider electrochemical reactions that occur near the triple phase boundary. However, if the electrochemical performance of SOFC electrodes is to be determined directly, in that case, discrepancies in TPB density in the reconstructions might affect the polarisation resistance [22]. The fact that the result of M^{slices} is closer to the M^{full} might be due to the reason that the 2D slices do not contain information about the double or TPB, as these descriptors are 3D quantities that are hard to reduce to two-dimensional information. Thus, the effect of a better approximation of the TPB in M^{slices} seems to be arbitrary. The density of the double phase boundary between nickel and YSZ is better captured in the reconstructions with the values of $\rho_{\text{DPB}}(M^{\text{orig}}) = 1.88 \mu\text{m}^{-1}$, $\rho_{\text{DPB}}(M^{\text{full}}) = 1.95 \mu\text{m}^{-1}$, and $\rho_{\text{DPB}}(M^{\text{slices}}) = 1.82 \mu\text{m}^{-1}$. This results in relative errors of up to 4%.

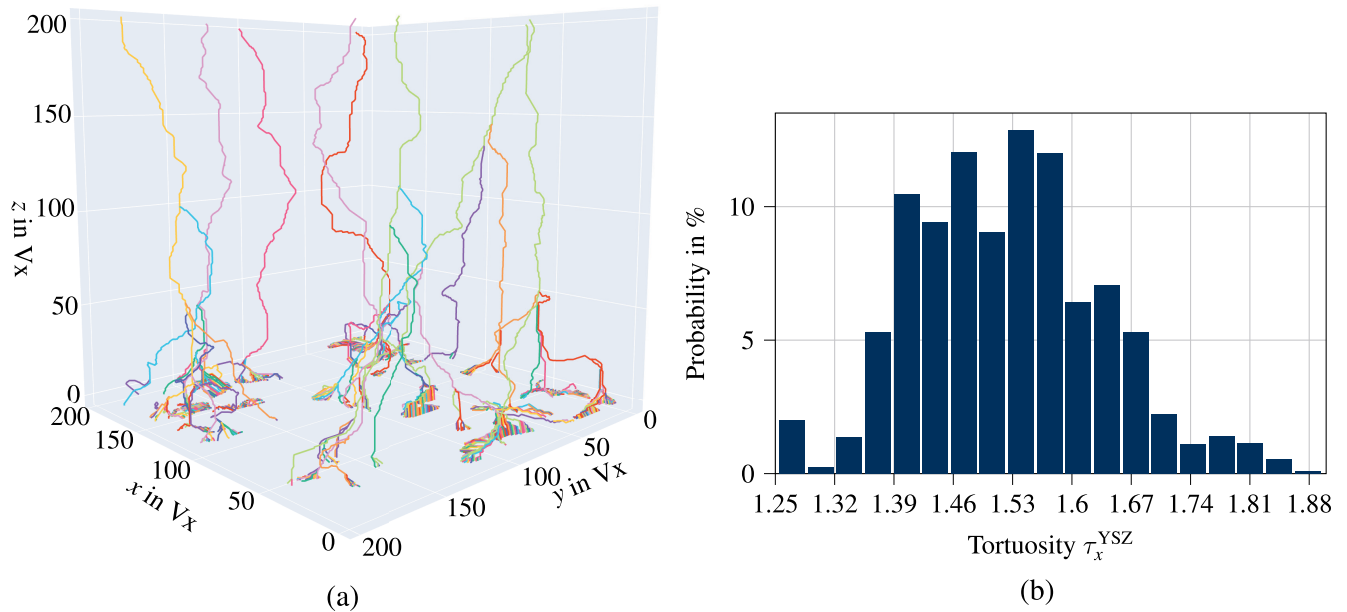


FIGURE 6 | Visualisation of (a) the shortest paths from $z = 0$ to $z = 200$ found with the Dijkstra algorithm in combination with the skeleton shortest-path method (SSPSM) and (b) the distribution of the tortuosity. (a) Paths from the surface at $z = 0$ to the opposite surface through an interconnected cluster of YSZ. (b) Tortuosity distribution of YSZ in z -direction.

The *tortuosities* of all microstructures are determined in terms of the skeleton tortuosity, as it can compute all percolated paths from one surface to the opposite surface in a reasonable time. The tortuosity is calculated for every path with Equation (8). The interconnected paths are exemplarily shown for the YSZ phase from surface $z = 0$ to the opposite surface, see Figure 6a. Thus, a distribution for the lengths of these paths can be extracted, see Figure 6b. However, in Figure 4b, only the mean tortuosities of each phase in each direction $\bar{\tau}_i^{(p)}$ are used for a comparison between the microstructures.

It is worth noting that the tortuosity is not used as a descriptor in the reconstruction process, as explained in Section 4.1. The differences of $\bar{\tau}_i^{(p)}$ between the original and the reconstructed microstructures can be clearly seen. The largest errors occur in the pore phase in all directions, which also contribute to the permeability differences in Section 5.2. For the other phases, the differences are lower, but the tortuosities of the reconstructed microstructures are still not in perfect agreement with the original microstructure. The deviating tortuosity values affect not only the permeability, but also the transport of ions and electrons. However, incorporating shortest-path algorithms into microstructure reconstruction poses two main challenges: differentiability and computational cost. For the differentiable step, existing shortest-path algorithms (e.g., Dijkstra) are difficult to adapt due to non-differentiable operations and iterative updates. Recent machine learning approaches address this by differentiating the algorithm's final output directly, using techniques such as linearisation or stochastic perturbations [72, 73]. For the Yeong-Torquato procedure, the high computational costs of repeated shortest-path evaluations remain a barrier, as efficient update rules—successfully developed for two-point correlations [66, 67]—are not yet available. Deriving such updates, potentially by tracking multiple shortest paths or developing heuristics to identify relevant pixel swaps, could substantially improve performance but requires significant algorithmic and data-structural innovations. These challenges highlight promising but technically demanding avenues for future research, particularly for applications like SOFCs, where connectivity information matters especially much.

5.2 | Physical Properties

The physical bulk properties of the components of the Ni-YSZ anode of the fuel cell, operated with pure hydrogen at a temperature of $T_{op} = 800^\circ\text{C}$, are used. These properties listed in Table 1 are employed in the homogenisation procedure to predict the effective physical properties, see Section 3.

TABLE 1 | Isotropic bulk properties of the constituents of the anode for each physical field at $T_{\text{op}} = 800^\circ\text{C}$.

	κ in $\text{W m}^{-2} \text{K}^{-1}$	λ^{e^-} in S m^{-1}	$\lambda^{O^{2-}}$ in S m^{-1}	μ in Pa s	E in MPa	G in MPa	α in —	A in $\text{MPa}^{-n} \text{s}^{-1}$	n in —
Ni	73.71	$2.13 \cdot 10^6$ [74]	—	—	$179.57 \cdot 10^3$ [75]	$69.23 \cdot 10^3$ [75]	$1.55 \cdot 10^{-5}$ [75]	$2.32 \cdot 10^{-14}$ [60]	4.6 [60]
YSZ	2.40	—	2.26 [76]	—	$113.59 \cdot 10^3$ [77]	$41.78 \cdot 10^3$ [77]	$1.08 \cdot 10^{-5}$ [77]	$3.71 \cdot 10^{-14}$ [60]	1.0 [60]
H ₂	0.18	—	—	$2.15 \cdot 10^{-5}$	—	—	—	—	—

Note: Please note that the values of physical quantities marked with “—” are negligible and were not considered during the numerical simulation.

TABLE 2 | Comparison of effective physical properties: Normed distance between the effective tensors with $\mathcal{E}^{\text{ph}}$ (Equation (38)) in %.

	κ^{eff}	$\lambda^{e^-, \text{eff}}$	$\lambda^{O^{2-}, \text{eff}}$	K^{eff}	C^{eff}	α^{eff}	M^{eff}
$\mathcal{E}^{\text{ph, full}}$ in %	2.62	5.01	6.76	11.80	2.08	3.77	5.35
$\mathcal{E}^{\text{ph, slices}}$ in %	6.67	10.37	8.99	14.62	3.25	6.87	4.95

In Table 2, the differences between effective physical properties computed from original and reconstructed microstructures are characterised by the relative error $\mathcal{E}^{\text{ph}}$ using the Frobenius norm:

$$\mathcal{E}^{\text{ph}} = \|(\cdot)_{\text{rec}} - (\cdot)_{\text{orig}}\|_F / \|(\cdot)_{\text{orig}}\|_F. \quad (38)$$

The values of each property are extracted from the numerical homogenisation procedure with linear boundary conditions (fluctuations set to zero), as detailed in Section 3. The numerical simulations were also performed using periodic boundary conditions, and the same qualitative trends could be recognised. The second-order tensors of the effective transport properties of the original microstructure M^{orig} at 800°C are shown in Equation (39):

$$\begin{aligned} \kappa^{\text{eff}} &= \begin{pmatrix} 10.5 & 0.2 & 0.3 \\ 0.2 & 11.1 & -0.6 \\ 0.3 & -0.6 & 13.4 \end{pmatrix} \text{W m}^{-1} \text{K}^{-1}, & \lambda^{O^{2-}, \text{eff}} &= \begin{pmatrix} 0.40 & 0 & 0 \\ 0 & 0.44 & -0.02 \\ 0 & -0.02 & 0.51 \end{pmatrix} \text{S m}^{-1}, \\ K^{\text{eff}} &= \begin{pmatrix} 2.3 & 0 & 0 \\ 0 & 2.4 & 0 \\ 0 & 0 & 2.9 \end{pmatrix} \cdot 10^{-3} \mu\text{m}^2, & \lambda^{e^-, \text{eff}} &= \begin{pmatrix} 2.1 & 0.1 & 0.1 \\ 0.1 & 2.3 & -0.2 \\ 0.1 & -0.2 & 2.9 \end{pmatrix} \cdot 10^5 \text{S m}^{-1}. \end{aligned} \quad (39)$$

It can be seen that the transport properties exhibit a slightly anisotropic behaviour. The material symmetries can be described by a dissipation potential, as it is shown in Langner et al. [24]. The relative errors, shown in Table 2, reveal that there is a very good agreement between the original and the reconstructed microstructures. The errors are higher for M^{slices} , as the geometric descriptors deviate slightly more than for M^{full} . The permeability K^{eff} differs the most, as the tortuosities for the pore phase also vary considerably in all directions, see Figure 4b. It is worth noting that the anisotropic behaviour is also captured by the reconstructions for all properties, see Figure 7 as an example of the effective elastic behaviour.

The effective elasticity tensor (written in the Voigt notation $[\cdot]^V$) and the effective thermal expansion tensor are calculated as:

$$[C^{\text{eff}}]^V = \begin{pmatrix} 82.80 & 30.86 & 33.04 & -0.60 & 0.22 & -0.43 \\ 30.86 & 84.81 & 33.40 & -1.63 & -0.01 & -0.46 \\ 33.04 & 33.40 & 96.90 & -2.03 & 0.23 & -0.26 \\ -0.60 & -1.63 & -2.03 & 28.54 & -0.11 & -0.02 \\ 0.22 & -0.01 & 0.23 & -0.11 & 28.44 & -0.44 \\ -0.43 & -0.46 & -0.26 & -0.02 & -0.44 & 26.45 \end{pmatrix} \text{GPa}, \quad \alpha^{\text{eff}} = \begin{pmatrix} 1.38 & -0.15 & 0 \\ -0.15 & 1.38 & 0.17 \\ 0 & 0.17 & 1.37 \end{pmatrix} \cdot 10^{-5} \text{K}^{-1}. \quad (40)$$

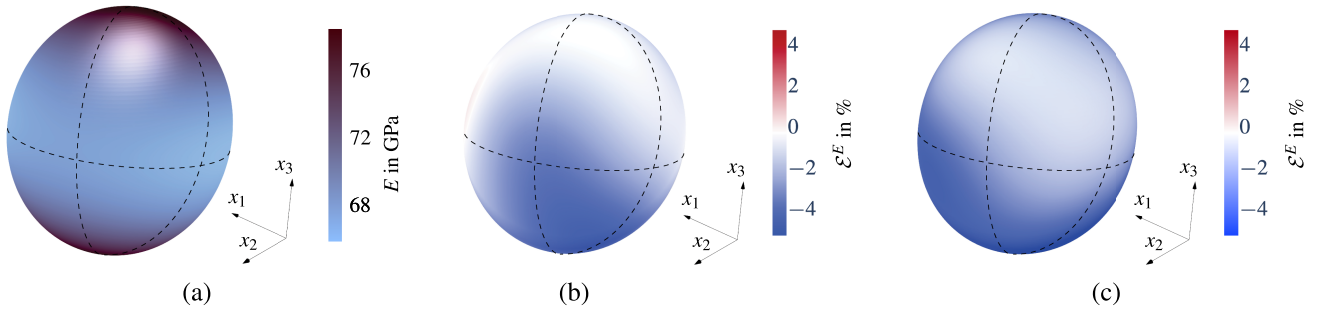


FIGURE 7 | Visualisation of the elastic surface to represent the purely elastic mechanical behaviour. The relative error (41) of the effective Young's modulus between the reconstructed and the original microstructures is shown. (a) Elastic surface of $\mathbf{M}^{\text{orig}}$; (b) Elastic surface of $\mathbf{M}^{\text{full}}$ and relative error to $\mathbf{M}^{\text{orig}}$; (c) Elastic surface of $\mathbf{M}^{\text{slices}}$ and relative error to $\mathbf{M}^{\text{orig}}$.

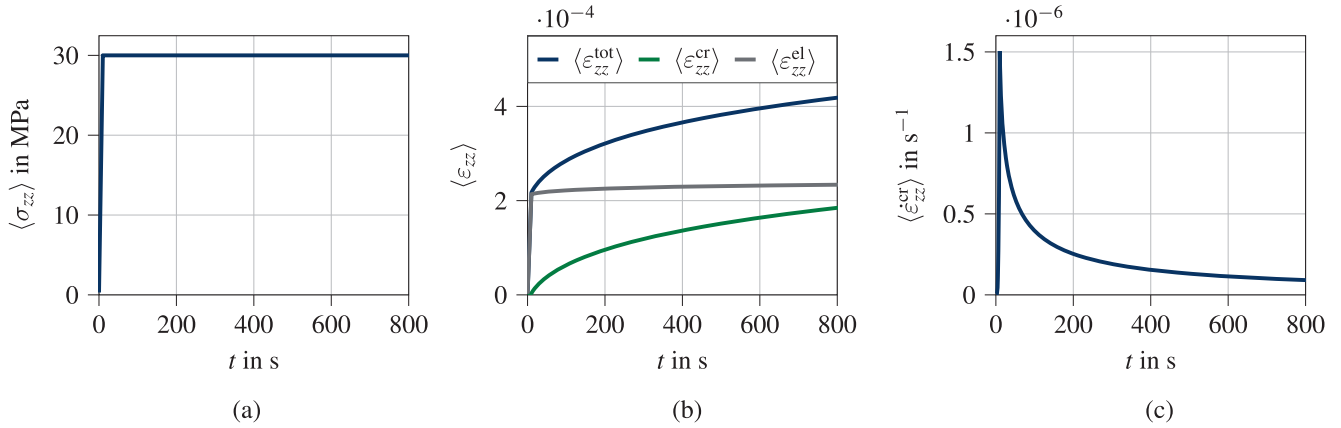


FIGURE 8 | Evolution of the averaged stress, the averaged strains and the averaged creep strain rate versus time for the original microstructure $\mathbf{M}^{\text{orig}}$ and an exemplary load case: tension in z-direction with a corresponding averaged stress of $\langle \sigma_{zz} \rangle = 30\text{MPa}$. The curves for the other load cases, for both the original and reconstructed microstructures, show the same qualitative course. (a) Average stress $\langle \sigma_{zz} \rangle$ in z-direction versus time: After applying a load, the stress is held constant; (b) Average total strain $\langle \epsilon_{zz}^{\text{tot}} \rangle$, elastic strain $\langle \epsilon_{zz}^{\text{el}} \rangle$ and creep strain $\langle \epsilon_{zz}^{\text{cr}} \rangle$ in z-direction versus time; (c) Average creep strain rate $\langle \dot{\epsilon}_{zz}^{\text{cr}} \rangle$ in z-direction versus time.

The effective elastic behaviour can be visualised by the elastic surface [78, 79], see Figure 7. Here, the Young's modulus E^{eff} is graphically represented to show the anisotropy of the material. The errors of E^{eff} between the reconstructed and the original microstructures are calculated by

$$\mathcal{E}^E = \frac{E^{\text{eff, rec}} - E^{\text{eff, orig}}}{E^{\text{eff, orig}}}. \quad (41)$$

To quantify the distance of the effective elasticity tensor to a certain material symmetry class, Equation (27) is used. Therefore, certain material symmetries (isotropic (*is*), cubic (*cu*), transversely isotropic (*ti*), orthotropic (*or*)) are assumed and the loss function is minimised, respectively. The losses are as follows: $\mathcal{L}^{(is)} = 12.21\text{GPa}$, $\mathcal{L}^{(cu)} = 11.59\text{GPa}$, $\mathcal{L}^{(ti)} = 1.69\text{GPa}$, $\mathcal{L}^{(or)} = 0.63\text{GPa}$. Thus, the transversely isotropic and orthotropic approximations are very close to the effective elastic behaviour. In the following, the closest approximation, the orthotropic material symmetry, was further used to compute the effective creep properties.

To determine the effective creep properties, multiple creep simulations are performed. For this purpose, the resolution of the image was reduced to 108^3 voxels to decrease the computational costs for the time-dependent simulations. The effect of the image resolution is estimated in the Appendix (Section A). It was found that the lower resolution has only a small influence on the effective mechanical behaviour. A Neumann-driven loading routine with three constant load magnitudes in *x*, *y* and *z* direction and one constant shear load in the *xy*, *xz* and *yz* planes is applied for a time of $t = 800\text{s}$. In Figure 8, the evolution of the averaged stress and the averaged strains is shown for an averaged stress applied in the

z-direction of $\langle \sigma_{zz} \rangle = 30 \text{ MPa}$. After a short ramp of loading the microstructure, the stress is kept constant (see Figure 8a) as it is also done in experimental creep tests. In Figure 8b, the average total strain $\langle \epsilon_{zz}^{\text{tot}} \rangle$, elastic strain $\langle \epsilon_{zz}^{\text{el}} \rangle$ and creep strain $\langle \epsilon_{zz}^{\text{cr}} \rangle$ versus time are shown. It can be seen that the total strain increases over time, since the internal variable, the creep strain, evolves. However, the rate of the creep strain decreases, which leads to a nearly linear increase of $\langle \epsilon_{zz}^{\text{cr}} \rangle$ at $t = 800 \text{ s}$. This is because the average creep strain rate $\langle \dot{\epsilon}_{zz}^{\text{cr}} \rangle$ decreases over time up to a constant level at the final time, see Figure 8c.

The averaged values of the creep strain rates and the stresses for all load cases after the last time increment are employed as an input for the optimisation problem (34) to get the effective creep strain values A^{eff} , n^{eff} and $\mathbb{M}^{\text{eff}}$. In this equation, the fourth-order tensor $\mathbb{M}^{\text{eff}}$ was chosen such that the equivalent stress equals the Hill stress [80] and thus, consists of six variables (F, G, H, L, M, N):

$$[\mathbb{M}^{\text{eff}}]^{\text{V}} = \begin{bmatrix} G+H & -H & -G & 0 & 0 & 0 \\ -H & H+F & -F & 0 & 0 & 0 \\ -G & -F & F+G & 0 & 0 & 0 \\ 0 & 0 & 0 & 2L & 0 & 0 \\ 0 & 0 & 0 & 0 & 2M & 0 \\ 0 & 0 & 0 & 0 & 0 & 2N \end{bmatrix}. \quad (42)$$

The Hill stress form of $\mathbb{M}^{\text{eff}}$ is chosen specifically because the pure elastic behaviour is almost orthotropic. If a different symmetry is recognised for elasticity in the material, other approaches to the form of the tensor are possible. As A^{eff} is supposed to be an independent parameter, one of the parameters F to N of the structural tensor $\mathbb{M}^{\text{eff}}$ needs to be constrained to enable the minimisation process. If we were to encounter isotropic creep behaviour, the sum of the parameters F, G , and H would equal 1.5, as $F = G = H = 0.5$. According to the equivalent von Mises stress and the isotropic creep analogy [81], it was determined that the relationship $F = 1.5 - G + H$ must be satisfied. Basin hopping [82], a global optimisation algorithm with local optimisation processes in each iteration, was found to achieve proper effective Norton creep parameters:

$$\begin{aligned} n^{\text{eff}}(\mathbf{M}^{\text{orig}}) &= 1.81, & n^{\text{eff}}(\mathbf{M}^{\text{full}}) &= 1.99, & n^{\text{eff}}(\mathbf{M}^{\text{slices}}) &= 2.08, \\ A^{\text{eff}}(\mathbf{M}^{\text{orig}}) &= 2.28e - 10 \text{ MPa}^{-n^{\text{eff}}} \text{ s}^{-1}, & A^{\text{eff}}(\mathbf{M}^{\text{full}}) &= 1.31e - 10 \text{ MPa}^{-n^{\text{eff}}} \text{ s}^{-1}, & A^{\text{eff}}(\mathbf{M}^{\text{slices}}) &= 9.70e - 11 \text{ MPa}^{-n^{\text{eff}}} \text{ s}^{-1}. \end{aligned}$$

The parameters A^{eff} and n^{eff} are very similar between the original and the reconstructed microstructures. Even though the exponent n^{eff} is slightly overestimated and the pre-factor A^{eff} is slightly underestimated for both reconstructions, the parameters agree very well with $\mathbf{M}^{\text{orig}}$. The directions of the creep, which is expressed by the fourth-order tensor $\mathbb{M}^{\text{eff}}$, show total relative errors between 4.95% and 5.35% according to the relative Frobenius norm (38), see Table 2. The parameters of $\mathbb{M}^{\text{eff}}$ are shown in Table 3. It can be seen that the anisotropic creep behaviour is properly captured by the reconstructed microstructures.

In general, the results of the physical properties show a very good agreement between the original and the reconstructed microstructures. However, the slight deviations originate from the differences in the geometrical properties. In particular, the transport properties strongly depend on the tortuosity of the respective phases, which is an important geometric descriptor that is not yet included in the microstructure reconstruction process. This highlights a possible improvement, that is, finding a differentiable approximation of the tortuosity and including it in the *MCRpy* tool. More investigations on specialised mutation rules or increasing the weight of ρ_{TPB} in the reconstruction process can achieve a stronger decrease of the triple phase boundary density within fewer iterations. In addition, further geometrical descriptors can be included, such as the continuous particle and pore size distribution, to account for bottleneck effects within each phase. This can

TABLE 3 | Parameters F, G, H, L, M, N of $\mathbb{M}^{\text{eff}}$ for the original and the reconstructed microstructures.

	F	G	H	L	M	N
$\mathbf{M}^{\text{orig}}$	0.42	0.48	0.60	1.12	1.01	1.10
$\mathbf{M}^{\text{full}}$	0.46	0.46	0.58	1.08	1.04	1.20
$\mathbf{M}^{\text{slices}}$	0.44	0.47	0.59	1.02	0.99	1.13

ensure an even more accurate description of the microstructure that leads to even better approximations of the physical properties of solid oxide fuel cell electrodes.

6 | Conclusion

This work represents a general framework to analyse electrodes of solid oxide fuel cells (SOFCs). Therefore, descriptors are defined that describe the composition and morphology of the microstructure. Based on these descriptors, realistic microstructures are reconstructed with a combination of the differentiable microstructure characterisation and reconstruction (DMCR) approach and the Yeong-Torquato algorithm. The effective physical properties that are determined by the underlying microstructure are numerically computed by the first-order homogenisation method. The effective transport properties (ionic, electronic, thermal conductivities, and permeability) and the mechanical properties (purely elastic, creep, thermal expansion) are calculated. The computational framework is applied to a Ni-YSZ anode obtained by focused ion beam-scanning electron microscopy (FIB-SEM). The results show very good agreement of geometrical and physical properties between the original and the reconstructed microstructures. It is even possible to generate the microstructure based on only three orthogonal slices, which would save tremendous experimental costs for the FIB-SEM procedure.

The framework can be used (i) to analyse the microstructure from a geometrical and physical perspective, followed by an implementation of these results into macroscopic models to have better predictions, (ii) to create a large microstructure-property database, and as a starting point (iii) for inverse designs of electrode microstructures. In particular, the presented simulation chain can be used to optimise the design of SOFC performance. Therefore, multiple microstructure samples that are either experimentally determined or artificially reconstructed by the current reconstruction method are needed to feed the microstructure-property database. In addition, the numerically computed effective properties need to be employed in macroscopic fuel cell models. At the fuel cell level, the maximum power and the mechanical stability can serve as key indicators that need to be maximised. By inverse design methods, it is possible to determine the most suitable microstructure design of electrodes for the best fuel cell performance. Thus, this work has the potential to contribute to the enhancement of the durability of SOFCs to more than 80,000 h, which is the target defined by the US Department of Energy [83]. Besides, the framework can also be used for electrodes of similar technologies, for example, SOFC electrolyzers and other types of fuel cells, such as the proton-exchange membrane fuel cell.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. M. A. Abdelkareem, K. Elsaid, T. Wilberforce, M. Kamil, E. T. Sayed, and A. Olabi, "Environmental Aspects of Fuel Cells: A Review," *Science of the Total Environment* 752 (2021): 141803, <https://doi.org/10.1016/j.scitotenv.2020.141803>.
2. E. V. Tsipis and V. V. Kharton, "Electrode Materials and Reaction Mechanisms in Solid Oxide Fuel Cells: A Brief Review: II. Electrochemical Behavior vs. Materials Science Aspects," *Journal of Solid State Electrochemistry* 12, no. 11 (2008): 1367–1391, <https://doi.org/10.1007/s10008-008-0611-6>.

3. S. J. McPhail, J. Kiviahio, and B. Conti, "The Yellow Pages of SOFC technology," *International Status of SOFC Deployment 2017* (2017): 1–51.
4. A. Choudhury, H. Chandra, and A. Arora, "Application of Solid Oxide Fuel Cell Technology for Power Generation-A Review," *Renewable and Sustainable Energy Reviews* 20 (2013): 430–442, <https://doi.org/10.1016/j.rser.2012.11.031>.
5. S. Zarabi Golkhatmi, M. I. Asghar, and P. D. Lund, "A Review on Solid Oxide Fuel Cell Durability: Latest Progress, Mechanisms, and Study Tools," *Renewable and Sustainable Energy Reviews* 161 (2022): 112339, <https://doi.org/10.1016/j.rser.2022.112339>.
6. J. Zheng, L. Xiao, M. Wu, et al., "Numerical Analysis of Thermal Stress for a Stack of Planar Solid Oxide Fuel Cells," *Energies* 15, no. 1 (2022): 343, <https://doi.org/10.3390/en15010343>.
7. X. Zhang, M. Wu, L. Xiao, et al., "Thermal Stress in Full-Size Solid Oxide Fuel Cell Stacks by Multi-Physics Modeling," *Energies* 17, no. 9 (2024): 2025, <https://doi.org/10.3390/en17092025>.
8. M. Peksen, D. Meric, A. Al-Masri, and D. Stolten, "A 3D Multiphysics Model and Its Experimental Validation for Predicting the Mixing and Combustion Characteristics of an Afterburner," *International Journal of Hydrogen Energy* 40, no. 30 (2015): 9462–9472, <https://doi.org/10.1016/j.ijhydene.2015.05.103>.
9. Z. Guo, X. Shi, Y. Chen, H. Chen, X. Peng, and P. Harrison, "Mechanical Modeling of Incompressible Particle-Reinforced Neo-Hookean Composites Based on Numerical Homogenization," *Mechanics of Materials* 70 (2014): 1–17, <https://doi.org/10.1016/j.mechmat.2013.11.004>.
10. Y. Luo, X. Chen, R. Liu, and W. Jiang, "Effects of a Cooling Channel on the Creep Damage and Failure Probability of Planar Solid Oxide Fuel Cells," *Frontiers in Materials* 10 (2023): 1238559, <https://doi.org/10.3389/fmats.2023.1238559>.
11. A. Semenov, E. Langner, M. El Hachemi, S. Belouettar, and T. Wallmersperger, "Modelling and Simulation of the Electro-Chemo-Thermo-Mechanical Behaviour of Solid Oxide Fuel Cells Considering Creep," *Acta Mechanica* (2025): 1–19, <https://doi.org/10.1007/s00707-025-04334-5>.
12. J. S. Cronin, J. R. Wilson, and S. A. Barnett, "Impact of Pore Microstructure Evolution on Polarization Resistance of Ni-Yttria-Stabilized Zirconia Fuel Cell Anodes," *Journal of Power Sources* 196, no. 5 (2011): 2640–2643, <https://doi.org/10.1016/j.jpowsour.2010.10.084>.
13. J. R. Wilson, J. S. Cronin, and S. A. Barnett, "Linking the Microstructure, Performance and Durability of Ni-Yttria-Stabilized Zirconia Solid Oxide Fuel Cell Anodes Using Three-Dimensional Focused Ion Beam-Scanning Electron Microscopy Imaging," *Scripta Materialia* 65, no. 2 (2011): 67–72, <https://doi.org/10.1016/j.scriptamat.2010.09.025>.
14. O. Pecho, O. Stenzel, B. Iwanschitz, et al., "3D Microstructure Effects in Ni-YSZ Anodes: Prediction of Effective Transport Properties and Optimization of Redox Stability," *Materials* 8, no. 9 (2015): 5554–5585, <https://doi.org/10.3390/ma8095265>.
15. J. Joos, "Microstructural Characterisation, Modelling and Simulation of Solid Oxide Fuel Cell Cathodes," (PhD Thesis, Karlsruher Institut für Technologie, Karlsruhe, 2016).
16. M. Guo and Z. Lin, "Long-Term Evolution of Mechanical Performance of Solid Oxide Fuel Cell Stack and the Underlying Mechanism," *International Journal of Hydrogen Energy* 46, no. 47 (2021): 24293–24304, <https://doi.org/10.1016/j.ijhydene.2021.04.196>.
17. H. Schlegel and R. Dawson, "Finite Element Analysis and Modelling of Thermal Stress in Solid Oxide Fuel Cells," *Proceedings of the Institution of Mechanical Engineers, Part A: Journal of Power and Energy* 231, no. 7 (2017): 654–665, <https://doi.org/10.1177/0957650917716269>.
18. L. Holzer and M. Cantoni, "Review of FIB-Tomography," in *559201222 of Oxford Series in Nanomanufacturing* (Oxford University Press, 2012), 410–435.
19. J. R. Wilson, W. Kobsiriphat, R. Mendoza, et al., "Three-Dimensional Reconstruction of a Solid-Oxide Fuel-Cell Anode," *Nature Materials* 5, no. 7 (2006): 541–544, <https://doi.org/10.1038/nmat1668>.
20. P. Shearing, J. Golbert, R. Chater, and N. Brandon, "3D Reconstruction of SOFC Anodes Using a Focused Ion Beam Lift-Out Technique," *Chemical Engineering Science* 64, no. 17 (2009): 3928–3933, <https://doi.org/10.1016/j.ces.2009.05.038>.
21. Z. Chen, X. Wang, F. Giuliani, and A. Atkinson, "Analyses of Microstructural and Elastic Properties of Porous SOFC Cathodes Based on Focused Ion Beam Tomography," *Journal of Power Sources* 273 (2015): 486–494, <https://doi.org/10.1016/j.jpowsour.2014.09.131>.
22. O. Pecho, A. Mai, B. Münch, T. Hocker, R. Flatt, and L. Holzer, "3D Microstructure Effects in Ni-YSZ Anodes: Influence of TPB Lengths on the Electrochemical Performance," *Materials* 8, no. 10 (2015): 7129–7144, <https://doi.org/10.3390/ma8105370>.
23. O. Stenzel, O. Pecho, L. Holzer, M. Neumann, and V. Schmidt, "Predicting Effective Conductivities Based on Geometric Microstructure Characteristics," *AIChE Journal* 62, no. 5 (2016): 1834–1843, <https://doi.org/10.1002/aic.15160>.
24. E. Langner, A. Semenov, A. Makradi, S. Gouttebroze, S. Belouettar, and T. Wallmersperger, "Macroscopic Properties of Solid Oxide Fuel Cell Electrodes via Microstructure-Based Numerical Homogenization," *PAMM* 24 (2024): e202400023, <https://doi.org/10.1002/pamm.202400023>.

25. E. Langner, H. Dehghani, M. El Hachemi, et al., "Physics-Based and Data-Driven Modelling and Simulation of Solid Oxide Fuel Cells," *International Journal of Hydrogen Energy* 96 (2024): 962–983, Elsevier.
26. G. Gaiselmann, M. Neumann, L. Holzer, T. Hocker, M. R. Prestat, and V. Schmidt, "Stochastic 3D Modeling of La_{0.6}Sr_{0.4}CoO_{3-δ} Cathodes Based on Structural Segmentation of FIB-SEM Images," *Computational Materials Science* 67 (2013): 48–62, <https://doi.org/10.1016/j.commatsci.2012.08.030>.
27. Q. Shao, A. Makradi, D. Fiorelli, et al., "Material Twin for Composite Material Microstructure Generation and Reconstruction," *Composites Part C: Open Access* 7 (2022): 100216, <https://doi.org/10.1016/j.jcomc.2021.100216>.
28. A. Gayon-Lombardo, L. Mosser, N. P. Brandon, and S. J. Cooper, "Pores for Thought: Generative Adversarial Networks for Stochastic Reconstruction of 3D Multi-Phase Electrode Microstructures With Periodic Boundaries," *NPJ Computational Materials* 6, no. 1 (2020): 82, <https://doi.org/10.1038/s41524-020-0340-7>.
29. T. Hsu, W. K. Epting, H. Kim, et al., "Microstructure Generation via Generative Adversarial Network for Heterogeneous, Topologically Complex 3D Materials," *Journal of Metals* 73 (2021): 90–102.
30. M. Baniassadi, H. Garmestani, D. Li, S. Ahzi, M. Khaleel, and X. Sun, "Three-Phase Solid Oxide Fuel Cell Anode Microstructure Realization Using Two-Point Correlation Functions," *Acta Materialia* 59, no. 1 (2011): 30–43, <https://doi.org/10.1016/j.actamat.2010.08.012>.
31. J. Feng, Q. Teng, X. He, L. Qing, and Y. Li, "Reconstruction of Three-Dimensional Heterogeneous Media From a Single Two-Dimensional Section via Co-Occurrence Correlation Function," *Computational Materials Science* 144 (2018): 181–192, <https://doi.org/10.1016/j.commatsci.2017.11.030>.
32. J. Fu, M. Wang, D. Xiao, S. Zhong, X. Ge, and B. Evans, "Hierarchical Reconstruction of 3D Well-Connected Porous Media From 2D Exemplars Using Statistics-Informed Neural Network," *Computer Methods in Applied Mechanics and Engineering* 410 (2023): 116049, <https://doi.org/10.1016/j.cma.2023.116049>.
33. J. Phan, M. Sarmad, L. Ruspini, G. Kiss, and F. Lindseth, "Generating 3D Images of Material Microstructures From a Single 2D Image: A Denoising Diffusion Approach," *Scientific Reports* 14, no. 1 (2024): 6498.
34. S. Buchanec, A. Sciazko, M. Mozdierz, and G. Brus, "A Novel Approach to the Optimization of a Solid Oxide Fuel Cell Anode Using Evolutionary Algorithms," *IEEE Access* 7 (2019): 34361–34372, <https://doi.org/10.1109/ACCESS.2019.2904327>.
35. P. Marmet, L. Holzer, T. Hocker, et al., "Stochastic Microstructure Modeling of SOC Electrodes Based on a Pluri-Gaussian Method," *Energy Advances* 2, no. 11 (2023): 1942–1967, <https://doi.org/10.1039/D3YA00332A>.
36. P. Seibert, M. Ambati, A. Raßloff, and M. Kästner, "Reconstructing Random Heterogeneous Media Through Differentiable Optimization," *Computational Materials Science* 196 (2021): 110455, <https://doi.org/10.1016/j.commatsci.2021.110455>.
37. P. Seibert, A. Raßloff, K. A. Kalina, et al., "Two-Stage 2D-To-3D Reconstruction of Realistic Microstructures: Implementation and Numerical Validation by Effective Properties," *Computer Methods in Applied Mechanics and Engineering* 412 (2023): 116098, <https://doi.org/10.1016/j.cma.2023.116098>.
38. P. Seibert, A. Raßloff, K. Kalina, M. Ambati, and M. Kästner, "Microstructure Characterization and Reconstruction in Python: MCRpy," *Integrating Materials and Manufacturing Innovation* 11, no. 3 (2022): 450–466, <https://doi.org/10.1007/s40192-022-00273-4>.
39. X. Fang and Z. Lin, "Numerical Study on the Mechanical Stress and Mechanical Failure of Planar Solid Oxide Fuel Cell," *Applied Energy* 229 (2018): 63–68, <https://doi.org/10.1016/j.apenergy.2018.07.077>.
40. S. Torquato, "Random Heterogeneous Materials," in *Interdisciplinary Applied Mathematics*, vol. 16 (Springer, 2002).
41. S. Torquato, "Statistical Description of Microstructures," *Annual Review of Materials Research* 32, no. 1 (2002): 77–111, <https://doi.org/10.1146/annurev.matsci.32.110101.155324>.
42. D. Chen, Z. Lin, H. Zhu, and R. J. Kee, "Percolation Theory to Predict Effective Properties of Solid Oxide Fuel-Cell Composite Electrodes," *Journal of Power Sources* 191, no. 2 (2009): 240–252, <https://doi.org/10.1016/j.jpowsour.2009.02.051>.
43. L. Holzer, P. Marmet, M. Fingerle, A. Wiegmann, M. Neumann, and V. Schmidt, *Tortuosity and Microstructure Effects in Porous Media: Classical Theories, Empirical Data and Modern Methods. Springer Series in Materials Science*, vol. 333 (Springer International Publishing, 2023).
44. J. Fu, H. R. Thomas, and C. Li, "Tortuosity of Porous Media: Image Analysis and Physical Simulation," *Earth-Science Reviews* 212 (2021): 103439, <https://doi.org/10.1016/j.earscirev.2020.103439>.
45. T. Y. Zhang and C. Y. Suen, "A Fast Parallel Algorithm for Thinning Digital Patterns," *Communications of the ACM* 27, no. 3 (1984): 236–239, <https://doi.org/10.1145/357994.358023>.
46. E. W. Dijkstra, "A Note on Two Problems in Connexion With Graphs," *Numerische Mathematik* 1, no. 1 (1959): 269–271, <https://doi.org/10.1007/BF01386390>.
47. H. Sumi, R. Kishida, J. Y. Kim, H. Muroyama, T. Matsui, and K. Eguchi, "Correlation Between Microstructural and Electrochemical Characteristics During Redox Cycles for Ni-YSZ Anode of SOFCs," *Journal of the Electrochemical Society* 157, no. 12 (2010): B1747, <https://doi.org/10.1149/1.3491345>.

48. L. Holzer, D. Wiedenmann, B. Münch, et al., "The Influence of Constrictivity on the Effective Transport Properties of Porous Layers in Electrolysis and Fuel Cells," *Journal of Materials Science* 48, no. 7 (2013): 2934–2952, <https://doi.org/10.1007/s10853-012-6968-z>.
49. M. Geers, V. Kouznetsova, and W. Brekelmans, "Multi-Scale Computational Homogenization: Trends and Challenges," *Journal of Computational and Applied Mathematics* 234, no. 7 (2010): 2175–2182, <https://doi.org/10.1016/j.cam.2009.08.077>.
50. S. Saeb, P. Steinmann, and A. Javili, "Aspects of Computational Homogenization at Finite Deformations: A Unifying Review From Reuss' to Voigt's Bound," *Applied Mechanics Reviews* 68, no. 5 (2016): 050801, <https://doi.org/10.1115/1.4034024>.
51. I. A. Baratta, J. P. Dean, J. S. Dokken, et al., "DOLFINx: The Next Generation FEniCS Problem Solving Environment," (2023), Preprint.
52. S. Balay, S. Abhyankar, M. Adams, et al., "PETSc Users Manual," in *PETSc Users Manual* (Argonne National Laboratory, 2019), 1–268.
53. I. Özdemir, W. A. M. Brekelmans, and M. G. D. Geers, "Computational Homogenization for Heat Conduction in Heterogeneous Solids," *International Journal for Numerical Methods in Engineering* 73, no. 2 (2008): 185–204, <https://doi.org/10.1002/nme.2068>.
54. J. Yvonnet, "Computational Homogenization of Heterogeneous Materials With Finite Elements," in *Solid Mechanics and Its Applications*, vol. 258 (Springer International Publishing, 2019).
55. S. Bishop, D. Marrocchelli, C. Chatzichristodoulou, et al., "Chemical Expansion: Implications for Electrochemical Energy Storage and Conversion Devices," *Annual Review of Materials Research* 44, no. 1 (2014): 205–239, <https://doi.org/10.1146/annurev-matsci-070813-113329>.
56. M. Muramatsu, K. Terada, T. Kawada, K. Yashiro, K. Takahashi, and S. Takase, "Characterization of Time-Varying Macroscopic Electro-Chemo-Mechanical Behavior of SOFC Subjected to Ni-Sintering in Cermet Microstructures," *Computational Mechanics* 56, no. 4 (2015): 653–676, <https://doi.org/10.1007/s00466-015-1193-7>.
57. H. G. Hahn, *Elastizitätstheorie: Grundlagen der linearen Theorie und Anwendungen auf eindimensionale, ebene und räumliche Probleme* (Vieweg+Teubner Verlag, 1985).
58. K. Naumenko, *Modeling of High-Temperature Creep for Structural Analysis Applications* (Martin Luther University Halle-Wittenberg, 2006).
59. F. H. Norton, *The Creep of Steel at High Temperatures*, No. 35 (McGraw-Hill Book Company, 1929).
60. K. Kwok, P. S. Jørgensen, and H. L. Frandsen, "Computation of Effective Steady-State Creep of Porous Ni-YSZ Composites With Reconstructed Microstructures," *Journal of the American Ceramic Society* 98, no. 9 (2015): 2873–2880, <https://doi.org/10.1111/jace.13712>.
61. A. Semenov, "Improvement of the Research Methods of Thermo-Elasto-Plastic Deformation Processes on the Base of Multimodel Analysis and the Exact Integration of Constitutive Equations (Russian)," (PhD Thesis, St. Petersburg University, St. Petersburg; 1996).
62. A. S. Semenov, A. C. Liskowsky, and H. Balke, "Return Mapping Algorithms and Consistent Tangent Operators in Ferroelectroelasticity," *International Journal for Numerical Methods in Engineering* 81, no. 10 (2010): 1298–1340.
63. M. Moakher and A. N. Norris, "The Closest Elastic Tensor of Arbitrary Symmetry to an Elasticity Tensor of Lower Symmetry," *Journal of Elasticity* 85, no. 3 (2006): 215–263, <https://doi.org/10.1007/s10659-006-9082-0>.
64. C. L. Y. Yeong and S. Torquato, "Reconstructing Random Media," *Physical Review E* 57, no. 1 (1998): 495–506, <https://doi.org/10.1103/PhysRevE.57.495>.
65. L. M. Pant, S. K. Mitra, and M. Secanell, "Stochastic Reconstruction Using Multiple Correlation Functions With Different-Phase-Neighbor-Based Pixel Selection," *Physical Review E* 90, no. 2 (2014): 023306, <https://doi.org/10.1103/PhysRevE.90.023306>.
66. M. G. Rozman and M. Utz, "Efficient Reconstruction of Multiphase Morphologies From Correlation Functions," *Physical Review E* 63, no. 6 (2001): 066701, <https://doi.org/10.1103/PhysRevE.63.066701>.
67. A. Adam, F. Wang, and X. Li, "Efficient Reconstruction and Validation of Heterogeneous Microstructures for Energy Applications," *International Journal of Energy Research* 46 (2022): 22757–22771, <https://doi.org/10.1002/er.8578>.
68. R. Bostanabad, Y. Zhang, X. Li, et al., "Computational Microstructure Characterization and Reconstruction: Review of the State-Of-The-Art Techniques," *Progress in Materials Science* 95 (2018): 1–41, <https://doi.org/10.1016/j.pmatsci.2018.01.005>.
69. P. Seibert, A. Raßloff, K. Kalina, et al., "On the Relevance of Descriptor Fidelity in Microstructure Reconstruction," *PAMM* 23 (2023): e202300116, <https://doi.org/10.1002/pamm.202300116>.
70. L. Holzer, O. Pecho, T. Hocker, B. Iwanschitz, and A. Mai, "FIB-Tomography Data of Ni-YSZ Anodes for Solid Oxide Fuel Cells (SOFC): Comparison of Pristine and Degraded Materials (Before/After Redox Cycling)," (2020), <https://zenodo.org/record/4056537>.
71. N. Shikazono, D. Kanno, K. Matsuzaki, H. Teshima, S. Sumino, and N. Kasagi, "Numerical Assessment of SOFC Anode Polarization Based on Three-Dimensional Model Microstructure Reconstructed From FIB-SEM Images," *Journal of the Electrochemical Society* 157, no. 5 (2010): B665, <https://doi.org/10.1149/1.3330568>.

72. M. Vlastelica, A. Paulus, V. Musil, G. Martius, and M. Rolínek, "Differentiation of Blackbox Combinatorial Solvers," (2019), arXiv Preprint arXiv:1912.02175.
73. Q. Berthet, M. Blondel, O. Teboul, M. Cuturi, J. P. Vert, and F. Bach, "Learning With Differentiable Pertubed Optimizers," *Advances in Neural Information Processing Systems* 33 (2020): 9508–9519.
74. U. Anselmi-Tamburini, "Electrical Properties of Ni/YSZ Cermets Obtained Through Combustion Synthesis," *Solid State Ionics* 110, no. 1–2 (1998): 35–43, [https://doi.org/10.1016/S0167-2738\(98\)00115-5](https://doi.org/10.1016/S0167-2738(98)00115-5).
75. V. E. Zinoviev, *Thermo-Physical Properties of Metals at High Temperatures* (Metallurgiya, 1989).
76. J. Ferguson, J. M. Fiard, and R. Herbin, "Three-Dimensional Numerical Simulation for Various Geometries of Solid Oxide Fuel Cells," *Journal of Power Sources* 58, no. 2 (1996): 109–122.
77. M. Radovic and E. Lara-Curzio, "Elastic Properties of Nickel-Based Anodes for Solid Oxide Fuel Cells as a Function of the Fraction of Reduced NiO," *Journal of the American Ceramic Society* 87, no. 12 (2004): 2242–2246, <https://doi.org/10.1111/j.1151-2916.2004.tb07499.x>.
78. T. Böhlke and C. Brüggemann, "Graphical Representation of the Generalized Hooke's Law," *Technische Mechanik* 21, no. 2 (2001): 145–158.
79. J. Nordmann, M. Aßmus, and H. Altenbach, "Visualising Elastic Anisotropy: Theoretical Background and Computational Implementation," *Continuum Mechanics and Thermodynamics* 30, no. 4 (2018): 689–708, <https://doi.org/10.1007/s00161-018-0635-9>.
80. R. Hill, "A Theory of the Yielding and Plastic Flow of Anisotropic Metals," *Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences* 193, no. 1033 (1948): 281–297, <https://doi.org/10.1098/rspa.1948.0045>.
81. R. von Mises, "Mechanik der festen Körper im plastisch-deformablen Zustand," *Nachrichten Von Der Gesellschaft Der Wissenschaften Zu Göttingen, Mathematisch-Physikalische Klasse* 1913 (1913): 582–592.
82. D. J. Wales and J. P. K. Doye, "Global Optimization by Basin-Hopping and the Lowest Energy Structures of Lennard-Jones Clusters Containing up to 110 Atoms," *Journal of Physical Chemistry A* 101, no. 28 (1997): 5111–5116, <https://doi.org/10.1021/jp970984n>.
83. U.S. Department of Energy, "DOE Technical Targets for Fuel Cell Systems for Stationary Combined Heat and Power," (2025), <https://energy.gov/eere/fuelcells/doe-technical-targets-fuel-cell-systems-stationary-combined-heat-and-power>.

Appendix A

Effect of Image Resolution on Creep

Here, the effect of the image resolution reduction on the transient creep behaviour is estimated on a smaller section of the microstructure. We used a microstructure of $300 \times 300 \times 300$ voxels and reduced the initial resolution by the standard (in this case) downsampling factor $1/3$ to 100^3 voxels. We compared the result with a downsampling factor of 0.18 , that is, a microstructure with 54^3 voxels. Creep simulations are conducted for various load directions, followed by an evaluation of the resulting homogenised creep properties. The effective creep properties of these microstructures are re-evaluated by the Norton parameters A^{eff} and n^{eff} and the structural tensor $\mathbb{M}^{\text{eff}}$. Suppose that the effective properties of the high-resolution microstructure could be used to predict the creep rate of the low-resolution microstructure. In that case, there is only a slight difference in resolution. Therefore, the abscissa in Figure A1 describes the effective creep rate of the microstructure with 54^3 voxels that results from the simulations. The predicted values that result from the Norton law with the optimised parameters of the microstructure with 100^3 voxels are plotted on the ordinate. If the optimised parameters of the microstructure perfectly fit the response of the low-resolution microstructure, all values would be on the 45° line.

As can be seen in Figure A1, the optimised creep parameters can accurately predict the response of the low-resolution microstructure. Thus, the differences between these microstructures are very small with respect to the time-dependent creep behaviour.

Measures for Anisotropy Differences for Transport Properties

Two additional measures can be employed to quantify the differences in anisotropy of the transport properties (i.e., conductivities and permeability) between the microstructures. Therefore, the eigenvalues Λ and eigenvectors of each second-order tensor are computed, and the maximum relative error $\mathcal{E}^{\Lambda, \max}$ between the eigenvalues for each transport behaviour is evaluated.

To quantify the similarity of the eigendirections of the transport tensors, their principal subspaces are compared using the concept of principal angles. The principal angles are obtained from the singular values of the matrix $\underline{M} = \underline{U}^T \underline{V}$ using a singular value decomposition (SVD), where $\underline{U}$ and $\underline{V}$ contain the orthonormal eigenvectors of the respective subspaces. Because two of the eigenvalues are often nearly degenerate, the associated eigenvectors are not uniquely defined; therefore, the comparison is performed at the level of the corresponding two-dimensional subspace. Within this work, the dominant eigenvectors and the degenerate subspaces can be consistently compared without ambiguity. This subspace-based approach provides a robust measure of the directional deviation between tensors, avoiding misleadingly large angles that may occur when nearly degenerate eigenvectors are assessed individually. Thus, three angles are reported in Table A1: the angle of the dominant eigenvector Θ_0 and the two principal angles Θ_1, Θ_2 of the 2D subspaces. The dominant eigenvector angle quantifies the deviation between the eigenvectors associated with the largest eigenvalue of the tensors,

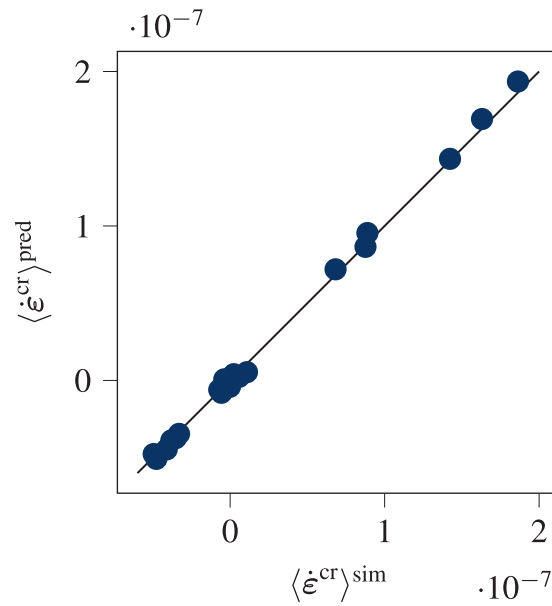
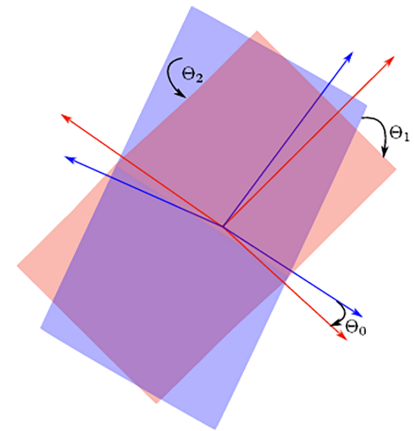


FIGURE A1 | Simulated averaged creep strain rates of the lower-resolution microstructure $\langle \dot{\epsilon}^{cr} \rangle^{sim}$ versus the prediction $\langle \dot{\epsilon}^{cr} \rangle^{pred}$ with the Norton model employing the effective creep parameters of the higher-resolution microstructure.

TABLE A1 | Metrics for anisotropy for transport properties

	κ^{eff}				$\lambda^{O^{2-},eff}$			
	$\mathcal{E}^{\Lambda,max}$	Θ_0	Θ_1	Θ_2	$\mathcal{E}^{\Lambda,max}$	Θ_0	Θ_1	Θ_2
$M^{orig} - M^{full}$	3.54 %	2.49°	1.40°	2.31°	6.70 %	1.58°	0°	1.58°
$M^{orig} - M^{slices}$	3.68 %	7.70°	0°	7.70°	10.03 %	5.83°	0°	5.83°

	$\lambda^{e,eff}$				K^{eff}			
	$\mathcal{E}^{\Lambda,max}$	Θ_0	Θ_1	Θ_2	$\mathcal{E}^{\Lambda,max}$	Θ_0	Θ_1	Θ_2
$M^{orig} - M^{full}$	5.79 %	5.64°	0.74°	5.65°	28.52 %	4.90°	0°	4.90°
$M^{orig} - M^{slices}$	3.15 %	8.04°	0°	8.04°	30.49 %	9.89°	0°	9.89°



Visualisation of principal angles

while the two subspace angles characterise the relative rotation of the plane spanned by the eigenvectors corresponding to the smaller eigenvalues.

The maximum relative differences of the eigenvalues are small for the conductivities, but high for the effective permeability. This is in agreement with the relative errors computed with the Frobenius norm. In addition, the angle of the dominant eigenvector and the two principal angles are computed for each physical field. It can be seen that only small deviations with respect to the anisotropic directions occur.